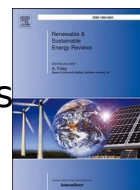




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Strategies for achieving carbon neutrality within the chemical industry

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ABSTRACT

This review paper explores the diverse strategies chemical companies are employing to achieve carbon neutrality amid escalating environmental regulations and societal demands for sustainability. The analysis focuses on the integration of advanced technologies, process optimization, and the transition to sustainable raw materials, with a special emphasis on the role of Carbon Capture, Utilization, and Storage (CCUS) technologies, renewable energy sources such as wind and solar, and the development of bio-based alternatives. These measures aim to lower the chemical sector's significant carbon footprint while also contributing to a broader sustainability goal. The evaluations of CCUS technology and renewable energy integration are central to this study's analysis, since they are critical for lowering direct CO₂ emissions and dependency on fossil fuels. The study also examines how adopting more sustainable raw materials and improving process efficiency might reduce manufacturing waste and carbon footprints. This article highlights top chemical companies that have made great strides toward their carbon neutrality objectives, such as those using CCUS technology to significantly reduce operating emissions, through successful case studies. It also addresses the challenges and opportunities facing the industry, including advances in technology, and evolving regulatory demands. The findings provide chemical companies with practical insights for achieving carbon neutrality and offer policymakers a roadmap for creating supportive frameworks. The study emphasizes how the chemical industry is at the forefront of sustainability practices and shows how corporate goals and environmental responsibility may work together to support global sustainability initiatives.

1. Introduction

1.1. Overview of the climate crisis and the need for carbon neutrality

As a result of growing industrialization and the consequent dependence on fossil fuels, concentrations of greenhouse gases, particularly carbon dioxide (CO₂), have increased dramatically, contributing to global warming and numerous of other ecological problems. According to G. Hu et al. CO₂ emissions from the combustion of fossil fuels comprised 88.6 % of total greenhouse gas emissions in 2019 [1]. The climate crisis highlights the urgent need for decarbonization across key sectors such as energy, transportation, industry, and building to stay within the 1.5– 2 °C global temperature increase limit set by the 2015 Paris Agreement [2–4]. Despite global initiatives, only 4.5 % of nations have achieved carbon neutrality, with the majority aiming to reach this goal between 2050 and 2070. Challenges persist in enhancing international collaboration for effective

adaptation and mitigation strategies [5]. On the policy front, governments around the world are rewarding CO₂ emissions reductions through mechanisms such as tax rebates and cap-and-trade systems, while simultaneously imposing economic disincentives such as significant carbon tax increases in countries like Norway and Canada. These initiatives aim to significantly reduce CO₂ emissions at refineries and chemical plants through increased energy efficiency, alternative fuels, and operational processes. However, larger reductions in carbon intensity will most certainly necessitate the use of carbon capture, utilization, and storage (CCUS) technology [6,7]. Recent advancements in CCS are centered on energy efficiency enhancement and minimization of environmental footprints in accordance with sustainability goals. Notably, Low-energy catalytic processes based on coordination have been shown to be a promising means of improving CO₂ capture efficiency without raising energy intensity or operating costs of conventional CCS processes. Low-energy catalytic processes use principles of coordination chemistry to enhance the selectivity of absorbents

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and their regeneration efficacy, resulting in low energy penalties and more environmentally friendly performance. According to recent studies, such as those discussed in Ref. [8], these advancements offer viable pathways for scaling CCS technologies in energy-intensive industries, including the chemical sector. With such innovations, CCS can be a more realistic and more sustainable path to carbon neutrality in the chemical industry. The International Energy Agency (IEA) has released a plan to achieve net-zero carbon emissions by 2050, highlighting key milestones such as tripling renewable energy capacity, doubling energy efficiency gains, and 75 % reductions in methane emissions from fossil fuels. The roadmap also emphasizes the need for all new heavy industry facilities to be near-zero-emissions capable by 2030 and sets ambitious targets for reducing CO₂ emissions by 80 % in advanced economies and 60 % in emerging economies by 2050. This transition involves ensuring that 90 % of power generation is derived from renewable sources, doubling nuclear energy capacity, and electrifying half of global energy consumption. However, it is also crucial to consider the impact of climate change on the viability and effectiveness of renewable energy sources. Studies, such as that in Ref. [9], show that climate change is likely to affect the generation potential of solar, wind, geothermal, hydroelectric, and biomass systems, as well as their availability. Temperature rise, changed precipitation patterns, and extreme weather events affect generation potential and increase operational challenges. Understanding these factors is critical for designing resilient and adaptive renewable energy strategies to achieve long-term carbon neutrality in the chemical industry. Furthermore, the plan includes the implementation of technologies to remove 1.7 gigatons (Gt) of CO₂ annually from the atmosphere [10]. In the Net Zero Emissions (NZE) Scenario, sharp reductions in CO₂ emissions are anticipated within the energy sector, dropping from 37 Gt in 2022 to 24 Gt by 2030—a reduction of about 35 %. By 2035, emissions are projected to decrease further to approximately 13.5 Gt, nearly 65 % below 2022 levels, driven by the scale-up of direct air capture and storage (DACs) and bioenergy with carbon capture and storage (BECCS), anticipated to reach around 0.6 Gt by 2035 and 1.7 Gt by 2050. By 2050, total CO₂ emissions from the energy sector are expected to achieve net zero, with residual emissions offset by removals through BECCS and DACs, excluding reliance on land-use offsets. Overall, greenhouse gas emissions from all sectors are expected to decrease by approximately 40 % by 2030 and 60 % by 2035. This strategic framework is designed to limit global temperature rise and mitigate climate change impacts, fostering collaboration among policymakers, industries, and stakeholders toward a sustainable, low-carbon future [11]. The transition to a renewable energy system is one of the key strategies to mitigate climate change across the globe. In combination with current approaches, new technologies like artificial intelligence (AI) are emerging to be excellent tools to facilitate such a transition. AI holds immense potential to make energy systems more efficient, for example, to make solar and wind energy more efficient, to predict energy demand patterns, and to stabilize energy grids. According to a new analysis [12], AI-based methods also enable more effective incorporation of renewable energy in power systems, predictive maintenance of energy systems, and optimization of resource utilization to manage challenges of intermittency and storage. The application of AI coupled with conventional renewable energy technologies is capable of providing more scalable, low-cost, and resilient cleaner energy technologies, hence a fundamental pillar of long-term decarbonization strategies.

1.2. Significance of chemical companies in global emissions

The chemical industry plays a pivotal role in global greenhouse gas (GHG) emissions and is essential in the global supply chain, notably segmented into upstream, midstream, and downstream sectors. The upstream sector focuses on producing basic petrochemical products and industrial gases; midstream involves the creation of intermediate chemicals; and downstream includes the manufacturing of final products such as plastics, pharmaceuticals, and synthetic rubbers. This diverse range of activities illustrates the extensive reach of the industry from basic chemical production to complex product development. Fig. 1 illustrates the global distribution of greenhouse gas (GHG) emissions by sector.

The industrial sector is the largest consumer of energy and the leading source of CO₂ emissions among all end-use sectors, accounting for 38 % of total final energy consumption and generating 47 % of CO₂ emissions, including those related to electricity and heat production [11]. Within the industrial sector, significant contributors include cement 27 %, iron and steel 25 %, and the chemical industry, which accounts for 18 % of emissions. The chemical industry's contribution is noteworthy, as it is responsible for a substantial portion of emissions, primarily due to energy-intensive processes and the production of essential materials such as ammonia and plastics [13,14]. Key energy-intensive industries, such as iron and steel, chemicals, non-metallic minerals, non-ferrous metals, and paper, consume almost 90 % of the coal used in industry, over 70 % of the oil, and approximately 55 % of the natural gas required for industrial activities [11]. Specifically, emissions from the chemical industry amounted to approximately 925 million metric tons of CO₂ in recent estimates [15]. The industry's significant use of fossil fuels as feedstocks embeds a substantial amount of carbon in the products manufactured. Projections suggest an increasing demand for chemical products such as fertilizers, plastics, and textiles, potentially leading to a significant rise in fossil fuel consumption for non-combusted feedstocks, which could notably increase emissions if current trends persist. The chemical industry's connection to the Paris Agreement's goals underscores the urgent shift towards sustainable and carbon-neutral operations. This transition not only focuses on enhancing the efficiency of existing infrastructure but also on developing innovative, low-carbon technologies. Significant technological innovation and economic investment are required to upgrade and implement energy-efficient systems, underlining the sector's crucial role in global emission reduction efforts. The chemical industry in Europe, particularly in Germany, is noted for its substantial contributions to CO₂ emissions, with rigorous targets set for substantial reductions by 2030. Similarly, industrial clusters across several European countries demonstrate significant potential for emission reductions through identified decarbonization levers such as steam generation and energy efficiency, which could reduce emissions by 50–60 %. In Japan, the chemical industry, accountable for a significant portion of national emissions, is pivotal in driving innovations aligned with Japan's 2050 carbon neutrality goal. This strategic positioning not only advances carbon neutrality but also sets a precedent for significant global CO₂ emission reductions [15,16].

1.3. Purpose and scope of the review paper

In this study, the literature review systematically explores a comprehensive set of strategies for achieving carbon neutrality within the chemical sector. The review adopts a holistic approach, covering all major sub-sectors of the chemical industry, including upstream (basic petrochemicals and industrial gases), midstream (intermediate chemicals), and downstream (final product manufacturing such as plastics, pharmaceuticals, and synthetic rubbers). While the review provides an industry-wide perspective, it also highlights sector-specific challenges and opportunities where relevant. The study aims to offer a comprehensive understanding of decarbonization strategies applicable to diverse chemical industry segments, ensuring relevance across the value chain. The study, beginning its initial literature search in December 2023 and updated in February

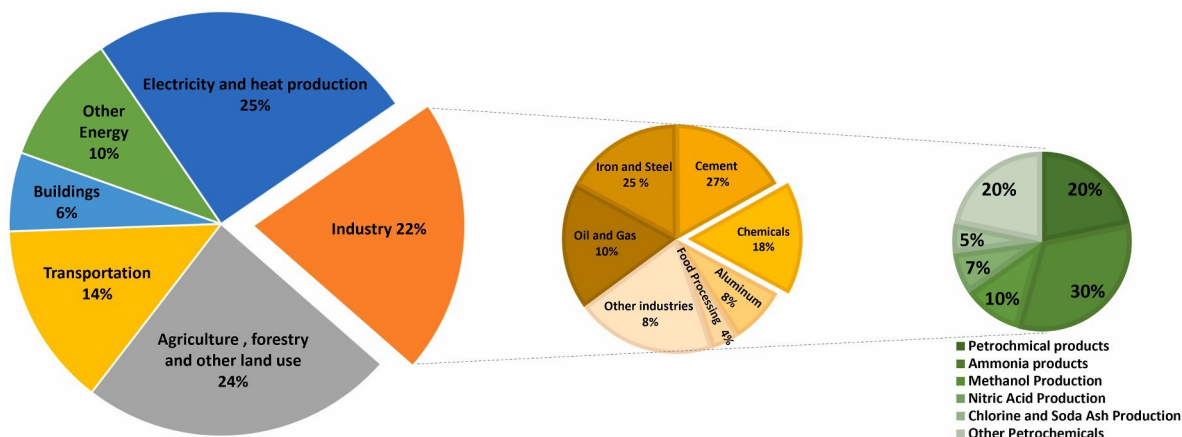


Fig. 1. Global GHGs emission by a: Sectors, b: Industries c: Chemical sector [10,11].

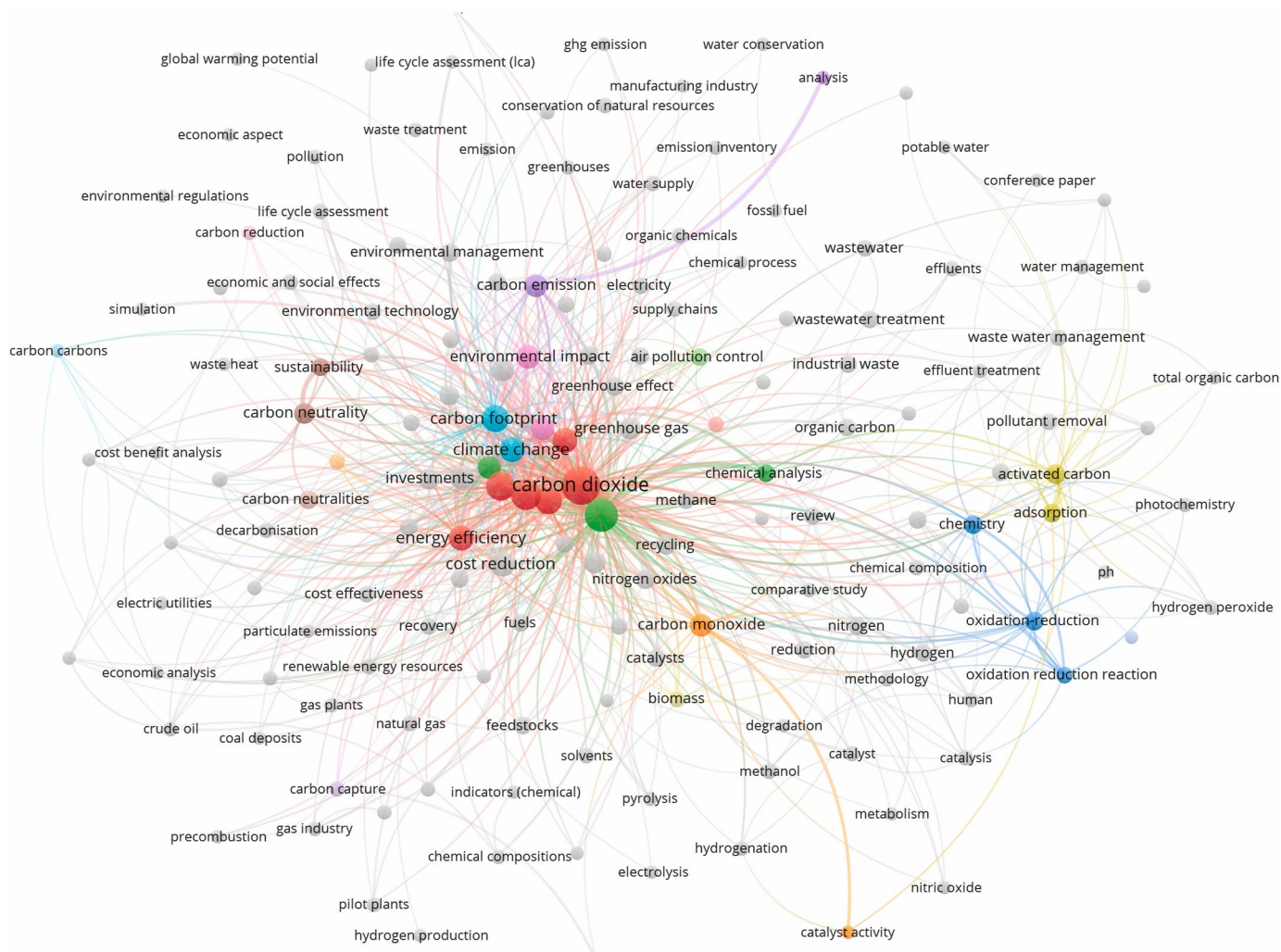


Fig. 2. Co-occurrence network of research themes related to carbon emissions in the chemical industry.

2025, employs a comprehensive bibliometric analysis, using databases including Scopus and Web of Science. This analysis identifies dominant themes and trends, with over 390 papers reviewed. Utilizing VOSviewer (Fig. 2) for keyword analysis highlights the central role of "carbon dioxide," which links to key themes such as "carbon footprint," "energy efficiency," and "recycling". These connections highlight the interdisciplinary nature of environmental strategies within the industry. The network also shows links to "renewable energy" and "sustainability," indicating a focus on sustainable practices and technologies. Economic aspects like "cost reduction" and "investments" are also depicted, emphasizing the financial dimensions of implementing environmental technologies.

The structure of this paper is as follows: Section 2 sets the stage by offering context and background information on the subject. Section 3 delves into both existing and emerging strategies for carbon reduction being considered and undertaken by chemical companies. Section 4 explores the obstacles and challenges tied to these strategies, while Section 5 addresses their environmental and economic impacts. Finally, Section 6 concludes with a summary of the findings and offers policy recommendations to support the transition to carbon neutrality within the chemical sector.

2. Background

2.1. Definition of carbon neutrality and its importance

Carbon neutrality implies net-zero CO₂ footprint, which can be offset by taking exactly the amount of CO₂ emitted into the atmosphere. This is made possible with the help of enhanced energy efficiency measures, utilization of renewable energy sources, afforestation, or deploying carbon capture and storage mechanisms. Addressing the climate crisis is critical across all sectors of human activity to counter long-term impacts such as extreme weather, rising sea levels, and biodiversity loss. Despite its critical importance, only a small percentage of nations have achieved carbon neutrality, aiming for targets between 2050 and 2070. A lack of international coordination has impeded the effectiveness of adaptation and mitigation strategies [5]. Projections indicate a decrease in CO₂ emissions by 18 % from 2022 levels by 2030, requiring a collaborative effort between corporate and public sectors to foster technological innovation, enhance efficiency, and boost recycling rates. This comprehensive approach underscores the global imperative to embrace carbon neutrality as a means to mitigate climate change impacts and secure a sustainable future [11].

2.2. The role of chemical companies in the global carbon footprint

Achieving carbon neutrality is essential for chemical companies given the substantial carbon emissions from their production processes. These processes often depend heavily on fossil fuels both as a source of energy and as raw materials, significantly enlarging the carbon footprint of the industry. Key high-temperature processes like steam cracking, reforming, and gasification, crucial for producing major chemicals such as ethylene, ammonia, and methanol, require extensive energy and emit considerable CO₂ and other greenhouse gases. These chemicals form the basis for a wide array of products including plastics, fertilizers, solvents, and pharmaceuticals [17]. In 2022, direct CO₂ emissions from primary chemical production were approximately 935 million tons, with a CO₂ intensity of around 1.3 tons per ton of product, indicating steady production levels [9]. The Net Zero Emissions (NZE) Scenario anticipates future disparities between CO₂ emissions and output levels,

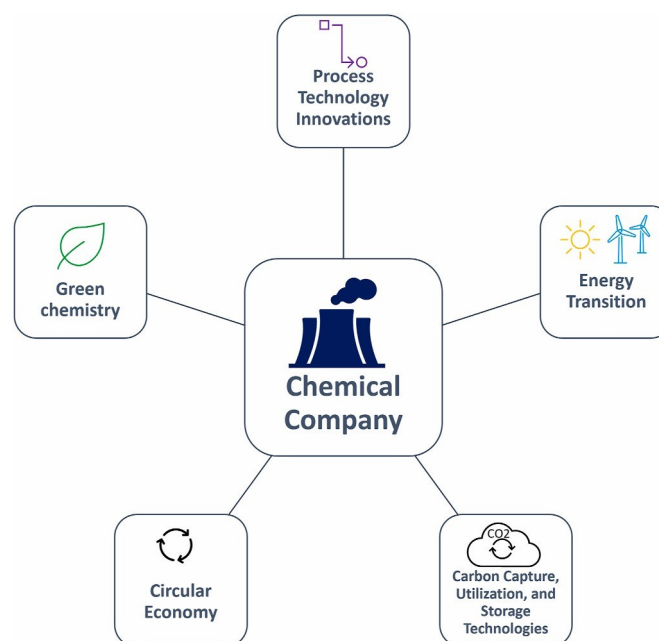
underlining the challenges of fully decarbonizing the industry. Nevertheless, substantial reductions in greenhouse gas emissions are achievable through enhanced carbon efficiency and adopting innovative emission-reducing technologies [18]. The sector's significant reliance on fossil fuels contributes notably to its greenhouse gas emissions, with electricity consumption accounting for about one-third of total emissions. This is particularly impactful in processes like electrolysis and medium-temperature heat treatments. Projected increases in electricity reliance—from 10 % of total energy use in 2019 to 25 % by 2070—will facilitate the adoption of renewable energy sources, potentially greatly reducing the sector's carbon footprint [19]. Base chemicals like petrochemicals (olefins and aromatics), methanol, and ammonia are major emission contributors, crucial for creating intermediate and downstream products. Production of these chemicals accounts for about one-third of the sector's total emissions, with additional emissions from feedstock production and further processing. Downstream processes, especially specialty chemical production involving smaller batches, tend to rely more on electricity and thus have relatively lower emissions [20]. To effectively manage and reduce emissions, the chemical industry must adopt innovative strategies that go beyond traditional carbon cycles. This includes converting CO₂ into useful commodity chemicals and implementing processes that circumvent CO₂ production entirely. According to the NZE scenario, CO₂ emissions from heavy industries are expected to decline by 20 % by 2030 and by 93 % by 2050, primarily due to enhanced efficiency, integration of advanced technologies, and improved material efficiency [10,11]. Comprehensive management of Scope 1, 2, and 3 greenhouse gas emissions, which includes identifying major sources, implementing reduction strategies, and continuous monitoring and reporting, is crucial for sustainability in the chemical sector. Strategies such as process electrification, adopting innovative catalysts for more efficient chemical production, improving energy efficiency, and utilizing green hydrogen are essential. Additionally, thorough accounting for Scope 3 emissions, which include indirect emissions throughout the value chain, is vital for a holistic environmental impact assessment, enabling the chemical sector to significantly contribute to global sustainability efforts [21]. Fig. 3 below displays the categories of upstream and downstream Scope 3 emissions, focusing on the categories that are most significant for chemical companies.

2.3. What is the state of the art in carbon reduction strategies for chemical companies, including new technologies?

According to the literature, the chemical industry has substantial opportunities to reduce its global carbon footprint through strategic initiatives. Innovations in process technology, such as the adoption of improved catalysis or electrochemical techniques, can help minimize energy consumption and emissions. Transitioning from fossil fuels to renewable energy sources like solar, wind, and bio-based alternatives is vital for reducing carbon emissions in chemical production. Implementing CCUS technologies at chemical facilities effectively captures CO₂ emissions before they can enter the atmosphere, sequestering them underground or converting them into valuable chemicals. Moreover, embracing a circular economy by increasing the recycling and reuse of chemical products and by-products reduces waste and decreases reliance on carbon-intensive materials. Employing green chemistry concepts to enhance processes and produce chemicals, such as bio-based solvents and biodegradable plastics, not only reduces the carbon

intensity of manufacturing processes but also aligns with the growing consumer demand for environmentally friendly products. These advancements represent a committed effort by the chemical sector to address environmental issues and meet global sustainability targets set by international organizations such as the IEA. Fig. 4 shows the important key strategies for reducing carbon footprints in chemical companies.

Recent advancements have significantly accelerated the deployment of clean technologies, yet the adoption of large-scale solutions like CCUS and liquid biofuels remains slow compared to smaller, more scalable technologies. This situation underscores the critical need for rapid development and broader implementation [11]. In sectors where direct electrification is not feasible, hydrogen presents a less carbon-intensive alternative, although it is not always cost-effective [22]. Policymakers and industry leaders must collaborate to establish market mechanisms and institutional policies that alleviate competitive disadvantages and facilitate necessary changes. Additionally, as the NZE scenario highlights a growing demand for low-emission hydrogen, particularly in heavy industry and transportation, there is an urgent need to increase production after 2030 to achieve long-term climate goals [23]. The projected decrease in emissions from the chemical sub-sector and the expected



GHGs						
CO_2	CH_4	N_2O	HFCs	PFCs	SF_6	NF_3
Scope 1 Emissions (Direct Emissions)		Scope 2 Emissions (Indirect Emissions from Electricity)			Scope 3 Emissions (Other Indirect Emissions)	
• Combustion Emissions • Process Emissions • Fugitive Emissions		• Electricity Usage • Heating and Cooling			• Upstream Activities • Downstream Activities • Waste Generated • Employee Commuting and Business Travel • Leased Assets and Franchises	

reduction in the share of fossil fuels in the energy mix by 2050 further emphasize the importance of transitioning to sustainable energy sources [24,25]. The transition to carbon-neutral chemical production is dependent on scalability and maturity of various decarbonization technologies. Deployment of renewable energy (solar, wind, and biomass) and energy efficiency technologies (digital monitoring, heat recovery, and process optimization) is already in commercial deployment (TRL 8–9), while other technologies are in their initial adoption phase [26]. CCS, particularly post-combustion capture, is already in use in large-scale plants but is hindered in terms of economics and infrastructure (TRL 6–7) [27]. Technologies used to produce hydrogen vary depending on the specific method employed. Conventional processes such as steam methane reforming are highly advanced and already in use widely, at TRL 9. However, green hydrogen production via water electrolysis based on renewable energy is at a lower TRL, typically between 6 and 7, indicating that it is in a phase of testing and demonstrations [28]. Technologies of DAC and electrified chemical processes

Fig. 3. Overview of Emissions across the Value Chain [16] CO_2 = Carbon dioxide; CH_4 = Methane; N_2O = Nitrous oxide; HFCs = Hydrofluorocarbons; PFCs =

Fig. 4. Strategic initiatives for reducing carbon footprints in chemical companies.

stand in their initial phase of studies and pilot demonstrations, that is, TRL 3–5. These technologies also raise significant challenges in terms of high energy demand and technical barriers to their progression to large-scale deployment. In order to facilitate their adoption in industry, there is a need to provide policy stimuli, infrastructure investments, and innovation. All these are crucial to moving such new technologies to higher TRLs and making them economically competitive in the chemical industry [29].

3. Existing carbon reduction strategies for chemical companies

The chemical sector, a significant industrial energy consumer, is ranked third in indirect CO_2 emissions among industry subsectors. This high ranking is primarily due to nearly half of its energy consumption being utilized as feedstock—fuel used as a raw material rather than for energy generation. Within the sector, ammonia production is the largest emitter, accounting for 45 % of total emissions. It is followed by methanol, contributing

28 %, and high-value chemicals, which account for 27 % of emissions [30,31]. Under the NZE scenario, it's projected that by 2030, existing technologies will account for about 80 % of emissions reductions in the global chemical industry. These technologies include recycling and reusing plastics, utilizing more efficient nitrogen fertilizers to decrease the demand for primary chemicals, and implementing various energy conservation measures. Post-2030, anticipated reductions are expected from the deployment of emerging technologies, such as specialized CCUS methods and the generation of electrolytic hydrogen using intermittent renewable energy sources [32]. Globally, chemical companies are adopting and developing carbon reduction techniques to tackle climate change. These range from established practices to innovative technologies aimed at decreasing the industry's carbon footprint. Strategies include enhancing energy efficiency, increasing electrification, using renewable energy, utilizing hydrogen fuels, harnessing bioenergy, and deploying CCUS. Their implementation is essential to advancing the circular economy, supporting sustainable production, and reducing lifecycle emissions [33]. Fig. 5 presents a systems-level framework integrating the five key carbon reduction strategies proposed in this study: energy-efficient processes, innovative renewable energy technologies, sustainable feedstocks, advanced carbon capture and utilization technologies (CCUS), and carbon offset programs and initiatives. These strategies are not isolated but interact to form a comprehensive decarbonization pathway. Energy-efficient

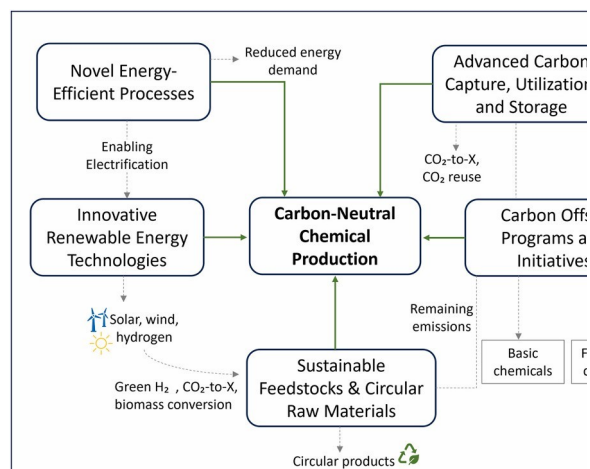


Fig. 5. Existing carbon reduction strategies for chemical companies. processes reduce demand, enabling cleaner electrification through renewable sources. Renewables, in turn, power hydrogen production and CO₂-to-X processes that feed into sustainable feedstock cycles. CCUS captures residual emissions for reuse or sequestration. Any remaining emissions, which cannot be eliminated by direct methods, are routed toward carbon offset programs to ensure full mitigation. The figure highlights the interdependencies and complementary nature of these strategies in supporting a robust and carbon-neutral chemical production system.

3.1. Carbon offset programs and initiatives

Carbon offset programs and initiatives within the chemical industry demonstrate a significant shift towards environmental responsibility and sustainability. These efforts are integral as the chemical sector is one of the largest contributors to global CO₂ emissions. A prominent approach in this sector is the adoption of carbon transformation strategies, commonly referred to as CO₂

to X. This process involves using CO₂, low-carbon hydrogen, and renewable energy sources as substitutes for traditional fossil fuels in chemical synthesis, promoting a significant reduction in carbon footprint [34]. These programs involve investing in environmental projects that reduce the emission of greenhouse gases in the atmosphere to balance out or "offset" the emissions made elsewhere, typically by industrial activities. Carbon offset schemes in the chemical industry include a number of ways for directly decreasing carbon emissions or compensating for them through external projects. For example, many chemical companies invest in renewable energy projects like wind and solar power, which provide clean energy sources to offset the emissions produced by the fossil fuels they use in their operations. Furthermore, reforestation and afforestation operations are prevalent since trees naturally absorb CO₂ from the atmosphere, substantially lowering net emissions. Additionally, the chemical industry's commitment to sustainable growth is further evidenced by significant investments in the carbon credit market. The "IETA Report on Trove Credit Investment" details that recent years have seen \$18 billion raised, predominantly by corporate investors, with a projected need for an additional \$90 billion by 2030 to meet the demands of a 1.5 °C scenario. This financial engagement supports a multitude of projects that not only aim to reduce emissions but also enhance biodiversity and promote sustainable development across the globe [35]. However, the carbon offset industry faces challenges such as credibility issues and the need for verifiable impact reduction. Advances in technology, particularly in digital monitoring and verification, offer potential to enhance transparency and accountability in the sector [36]. The International Economics Institute's 2023 report examines how carbon pricing mechanisms are encouraging the chemical industry to reduce emissions by making carbon emissions more expensive, pushing companies toward carbon reduction innovations or investments in carbon offset projects. Additionally, the industry's adaptation to global regulations reflects its ability to align with evolving environmental policies [37].

3.2. Novel energy-efficient processes

Incorporating cutting-edge energy-efficient processes into chemical companies is a comprehensive strategy aimed at reducing environmental impact while maintaining or enhancing production efficiency. This multifaceted approach emphasizes innovation, optimization, and sustainability. By integrating advanced carbon reduction technologies and modern energy-efficient methods, a chemical company can significantly lower its carbon footprint, improve energy efficiency, and transition to more sustainable operations. The implementation of these advanced energy-efficient processes not only helps in reducing the carbon footprint but also aligns with global sustainability goals. Adopting these technologies is crucial to transition towards more sustainable and environmentally friendly operations. This approach includes several key elements will be explained in detail in the following sections.

3.2.1. Waste heat recovery and utilization

Implementing waste heat capture and reuse technologies in chemical processes can greatly enhance energy efficiency, reduce operational costs, and lower greenhouse gas emissions. This approach involves deploying various technologies and methodologies tailored to the specific needs and conditions of the chemical processing facility. Techniques such as heat exchangers and cogeneration systems can recover heat for use in other processes, thus reducing the plant's overall energy

consumption [33]. Chemical companies are structured hierarchically, encompassing sites, processes, and operational units. Each level presents opportunities for energy integration through various strategies and procedures. Waste heat is generated at various stages of chemical processing within a chemical plant, and each recovery method differs, as illustrated in Fig. 6 [38].

In recent years, there has been significant growth in efforts to optimize the use of waste heat, particularly low-temperature and latent heat. This has spurred extensive research across various scales, from individual processes to entire sites, and even across multiple diverse locations [39]. A ranking criterion for designing waste heat recovery systems evaluates the energy efficiency of different technologies and their capacity to reduce greenhouse gas emissions. This structured approach ensures that the most effective and sustainable solutions are implemented to enhance overall operational efficiency and environmental performance [40,41]. Significant focus is placed on converting and utilizing low-temperature or low-potential waste heat at the process level. Organic Rankine Cycles (ORCs) are commonly employed to recover low-temperature waste heat, leading to substantial energy savings [42]. In refineries, substantial waste heat must be cooled to ambient temperatures, with some waste heat reaching temperatures as low as the surrounding environment. The effectiveness of an ORC system in recovering this low-temperature waste heat is highly dependent on the choice of working fluid and the operating parameters used [43].have developed an innovative method for simultaneously selecting the working fluid and determining the operational conditions for an ORC system. This approach is based on a new parameter that accurately predicts the pinch point between the waste heat source and the working fluid. It allows for precise calculation of the working fluid's flow rate and the amount of heat recovered, thus identifying the optimal working fluid and its corresponding operational conditions.

3.2.2. Process intensification

To enhance the efficiency and environmental sustainability of chemical processes, process intensification approaches are being emphasized. These methods include micro-reactor technology and advanced catalysis, which significantly boost reaction rates and selectivity while reducing energy consumption. Advanced catalysts accelerate and selectively drive chemical reactions, thereby lowering the energy required for production [44]. These catalysts are crucial for improving process efficiency and selectivity, reducing energy consumption, and minimizing the production of unwanted byproducts. Modern catalyst technology is transforming chemical processes through a number of revolutionary improvements. Nano-scale engineering improves catalysts by increasing surface area and introducing

unique active sites, hence increasing reactivity and specificity. Enzymatic and biomimetic catalysts are designed to mimic natural processes, resulting in increased efficiency and selectivity under milder circumstances, reducing reliance on harsh chemicals and extreme conditions [45]. Furthermore, advances in heterogeneous catalysts are making them easier to separate and reuse, resulting in waste reduction and cheaper operating costs. Collectively, these innovations are pushing the boundaries of catalyst technology, resulting in more sustainable and economically viable chemical production processes [46]. Microreactors, characterized by their small size and high surface area-to-volume ratio, offer precise control over reaction conditions, which improves efficiency and minimizes waste. These processes provide numerous benefits, including superior heat and mass transfer, reduced energy consumption, and enhanced safety. By integrating microreactors, chemical companies can achieve higher reaction efficiencies and lower energy requirements, contributing to a significant reduction in carbon emissions. For example, in chemical synthesis, microreactors ensure more efficient heat management and faster reaction times, leading to considerable energy savings [47]. Additionally, there is a growing focus on heat recovery in these processes, achieved through combined intensification and integration techniques, further enhancing overall process efficiency [48]. [49]have developed a novel energy-saving process configuration for dimethyl ether (DME) production using self-heat recuperation technology. This approach significantly reduces energy consumption in both thermal and separation processes. While DME is typically produced either indirectly from methanol or directly from syngas, their method shows up to an 80 % reduction in thermal utility demands, though it requires some power usage [50]. conducted research on the utilization of process intensification methods for increasing dimethyl ether production, employing integrating units such as dividing-wall columns and reactive distillation. These technologies have the potential to save up to 30 % and 40 % on capital and operational expenditures, respectively. In addition, the study emphasizes alternative methodologies that combine chemical reactions, separation, and recovery processes into a single unit, resulting in significant energy savings and reduced equipment requirements when compared to standard methods.

3.2.3. Electrification of chemical processes

Electrification of chemical processes refers to the shift from using traditional fossil fuel-based energy sources to electricity for powering chemical reactions and production techniques. This transformation is primarily driven by the need to reduce greenhouse gas emissions and increase energy efficiency in the chemical industry. With the adoption of renewable energy sources like solar and wind, the decarbonization of electricity has

Distillation Processes	During distillation, significant heat used to vaporize components allows for the capture of exhaust heat from both the hot distillate and the condenser's cooling water.
Reaction Processes	In reaction processes, waste heat from cooling reaction vessels or exiting product streams can be effectively reused for other purposes.
Heat Exchange Systems	Waste heat from hot streams exiting reactors, heaters, or furnaces can be captured and transferred to incoming feed streams or other heat-requiring processes.
Steam Generation and Boiler Systems	Waste heat from boiler exhaust gases can be harnessed via heat exchangers or by preheating boiler feedwater to enhance overall energy efficiency.
Cooling Systems	Waste heat from hot cooling water used in chemical process cooling systems can be extracted before discharge and repurposed for heating.
Ventilation and Exhaust Systems	Waste heat from exhaust gases of equipment like pumps, compressors, and reactors can be reclaimed with heat exchangers or used for space heating or preheating incoming air.
Flue Gas and Combustion Processes	Waste heat from furnace flue gases can be captured using heat recovery steam generators (HRSG) or Organic Rankine cycles (ORC) for power generation or process heating.

Fig. 6. Waste heat capture and reuse technologies in chemical processes.

advanced, contributing significantly to a sustainable future. Given the substantial energy consumption and high greenhouse gas emissions from the commodity chemical industry, its shift toward electrification could dramatically reduce global carbon dioxide emissions [51]. The global demand for electricity is expected to more than double from 2020 to 2050, with the industrial sector, including chemicals, experiencing the most significant increase, surging by over 11,000 TWh [32]. As renewable electricity becomes more affordable and accessible, it is increasingly used for processes like electrolysis and providing process heat necessary for transforming primary chemicals into intermediate and final products. Moreover, ongoing research aims to apply this cleaner energy source in steam cracking processes, essential for producing high-value chemicals, although commercial implementation is pending [19]. Currently, hydrocarbon steam cracking, known for its cost-efficiency and reliability, consumes significant energy and emits about 300 million tons of CO₂ annually [52]. Transitioning from gas-fired to electric furnaces could significantly reduce these emissions, with many chemical companies starting to adopt electric heating solutions to move away from combustion-based heat generation [53]. Electric cracking, or e-cracking as shown in Fig. 7, is a cutting-edge technology that employs electricity to split hydrocarbons into smaller molecules. Unlike traditional cracking methods that rely on fossil fuels for heat, generating substantial CO₂ emissions, e-cracking can leverage renewable electricity, drastically lowering the carbon footprint of the cracking process. This development represents a significant step forward in the pursuit of carbon-neutral chemical manufacturing [54].

Significant investments are flowing into transformative manufacturing technologies aimed at cleaner facilities by 2050. One such technology, Fluidized Catalytic Dehydrogenation (FCDh), enhances the efficiency of key processes such as the production of olefins like propylene. By using fluidized catalyst beds, FCDh optimizes heat and mass transfer, which accelerates reaction rates and improves conversion efficiencies, resulting in significant energy savings and emission reductions. This contributes to more sustainable chemical manufacturing practices [55]. Similarly, Ethane Dehydrogenation (EDH) technology, which transforms ethane into ethylene—an essential chemical industry component—offers a more energy-efficient alternative to the traditional steam cracking process. By employing catalysts that operate at lower temperatures and pressures, EDH reduces greenhouse gas emissions and operational costs, providing a more sustainable method for ethylene production. These advancements not only align with environmental goals by reducing the carbon footprint but also lower operational costs, promoting industry-wide adoption of cleaner technologies [56].

3.2.4. Olefin production and its role in the carbon cycle

Olefins, also known as alkenes, such as ethylene and propylene, are essential components of the petrochemical industry and contain at least one carbon-carbon double bond.

Olefins are traditionally manufactured using steam cracking, a method in which hydrocarbons such as ethane, propane, and naphtha are heated to high temperatures and broken down into smaller molecules. Other routes include catalytic cracking and dehydrogenation, which utilize catalysts to lower the temperature at which breakdown takes place and hence perhaps offer routes to more energy-efficient production. The chemical industry is addressing carbon emissions in olefin production through six main strategies, as illustrated in Fig. 8: using traditional resources like coal, oil, and gas, which are high in emissions and require CCS to reduce their impact; biomass, which converts renewable biological materials into olefins but is limited by resource availability; recycling plastics to transform waste into valuable chemicals; and both CCS and carbon capture and utilization (CCU) to either store or reuse captured CO₂. These approaches vary in sustainability and effectiveness, highlighting the complexity of achieving a reduced carbon footprint in olefin production. Each pathway offers unique benefits but also faces distinct challenges, emphasizing the need for an integrated and versatile strategy to effectively lower emissions [57].

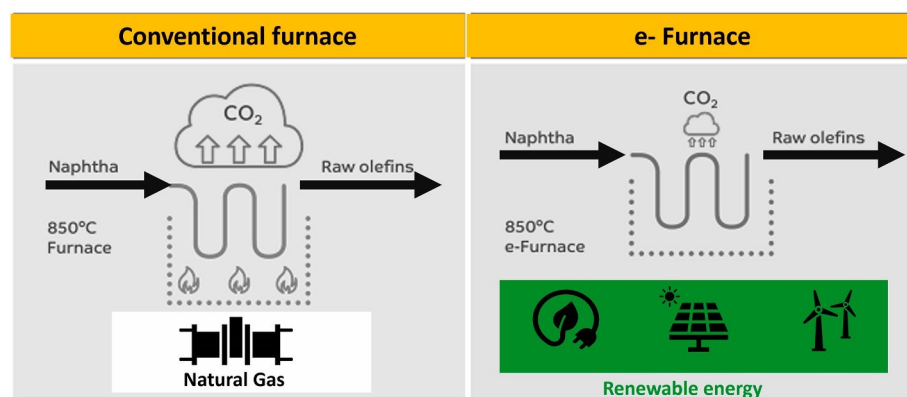


Fig. 7. Two Heating Concepts (Conventional and e Furnace).

Despite advances in technology that can minimize the carbon

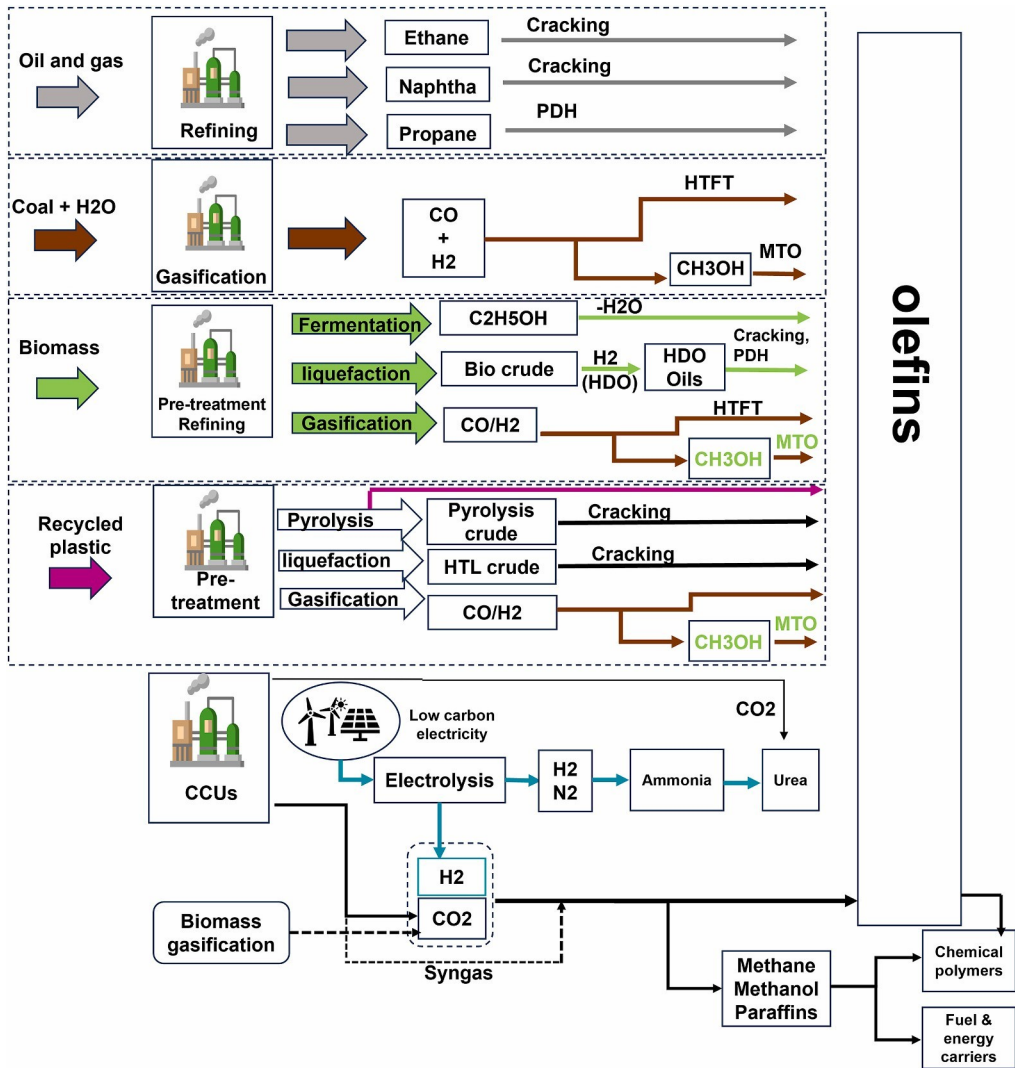


Fig. 8. Olefins production routes (reproduced from Ref. [50]) (propane dehydrogenation (PDH), High Temperature Fischer- Tropsch (HTFT), hydrodeoxygenation (HDO), propane dehydrogenation (PDH), methanol-to-olefins (MTO)).

footprint of olefin production, traditional fossil-based methods are expected to remain predominant for the next 2-3 decades. Techniques like Best Available Techniques (BAT) in naphtha cracking could reduce emissions by 20 %, and switching to ethanol as a feedstock could cut emissions by up to 45 %. Economic barriers, however, may slow the adoption of these greener solutions. The scarcity of sustainable feedstock sources means that the use of green olefins will increase only gradually, supported by advances in biomass gasification and recycling plastic waste. Additionally, Power-to-X technologies promise dramatic emissions reductions of 80–90 %, though their cost-effectiveness depends significantly on regulatory incentives [57]. A techno-economic analysis highlights that while traditional naphtha cracking is more cost-effective due to lower feedstock costs, the oxidative coupling of methane (OCM) method, though costlier, incorporates effective CO₂ capture. Despite the challenges, olefins play a vital role in sustainable development across sectors like energy, food security, and transportation [58].

3.2.5. *Smart grid technologies and demand response in the chemical industry*

Through increased energy efficiency and decreased carbon footprints, digital technologies like demand response systems and smart grids are transforming industrial operations in the chemical industry. Real-time energy management is made possible by smart grids, and demand response systems modify use in response to changes in energy prices and availability [59]. Energy Management Systems (EMS) and Advanced Metering Infrastructure (AMI) are two examples of technologies that offer comprehensive analytics to optimize energy use and facilitate the incorporation of renewable energy sources. Chemical facilities can also use incentives to reduce costs and emissions by modulating energy during peak hours through Automated Demand Response (ADR) systems. In addition to increasing operational effectiveness, this strategic use of digital tools helps the industry achieve its environmental goals [60,61].

3.3. *Innovative renewable energy technologies*

Renewable energy consumption plays a crucial role in carbon emissions, with a positive long-term correlation between the two and a limited short-term impact of energy consumption on emissions. In the chemical industry, various innovative renewable energy technologies are crucial for reducing carbon footprints [62]. These advancements allow the industry to shift from reliance on fossil fuels to more sustainable and eco-friendly energy sources. Technologies like Advanced BECCS mark significant advancements, transforming bioenergy production by integrating carbon capture to sequester CO₂ during energy production. This helps achieve negative carbon emissions, vital for meeting net-zero targets and beneficial for carbon-intensive industries like chemicals [63]. Floating solar panels offer another innovative solution, using water bodies to install photovoltaic systems that conserve land and reduce water evaporation, while also increasing renewable energy capacity and enhancing the efficiency of solar panels due to the cooling effects of water [64]. Additionally, High-altitude wind power utilizes stronger and more consistent winds at higher elevations using innovative airborne wind energy systems. This method surpasses traditional ground-based turbines by harnessing more powerful and reliable wind currents. For the chemical industry, integrating high-altitude wind power can transform energy sourcing by providing a more stable and plentiful wind energy supply. This technology boosts both the efficiency and sustainability of energy use in the sector and could fundamentally change renewable energy

strategies across multiple industries [65]. Nano-enhanced solar cells mark a significant advancement in photovoltaic technology by using nanotechnology to boost efficiency and reduce costs. Incorporating materials like perovskites and quantum dots, these cells achieve higher efficiencies and lower production costs than traditional solar technologies. This breakthrough could make solar energy more accessible and cost-effective for energy-intensive sectors such as the chemical industry, enhancing sustainability and economic viability in energy practices. This innovation represents a major progression in developing renewable energy solutions that cater to the needs of high-consumption industries [66]. Artificial photosynthesis replicates natural photosynthesis by using solar energy to convert CO₂ and water into carbohydrates and oxygen, offering a pathway to sustainable fuel and chemical production [67]. This method marks a transformative shift towards sustainable chemical production by utilizing solar energy to convert CO₂ and water into valuable petrochemicals like olefin. It begins with photolytic water splitting to produce hydrogen, followed by hydrogen separation through specialized membranes, and culminates in a catalytic reaction where hydrogen is combined with industrial CO₂ emissions. This process not only reduces greenhouse gas emissions but also contributes to a carbon-neutral chemical industry, showcasing a significant advancement in utilizing solar energy for direct CO₂ conversion [68]. Solid-state batteries, utilizing solid electrolytes in place of traditional liquid ones, deliver enhanced energy storage solutions, pivotal for the chemical industry's shift towards sustainable operations. These batteries significantly advance energy storage technology by offering safer, higher energy density options, leading to more compact and robust power storage. This improvement is crucial for effectively harnessing intermittent renewable energy sources within the chemical sector [69]. Thermophotovoltaic (TPV) cells, converting thermal energy directly into electrical energy, promise enhanced energy conversion efficiency, transforming waste heat from chemical processes into valuable electricity [70]. While these technologies provide promising pathways for reducing emissions, their integration faces challenges such as achieving technological maturity, cost-effectiveness, scalability, and compatibility with existing infrastructure [71].

One major issue is intermittency, as the availability of solar and wind energy fluctuates due to weather conditions, leading to supply variability [72]. This inconsistency can be a challenge to grid stability, particularly when penetration of renewables is high. The other primary challenge is energy storage, in that existing technologies such as lithium-ion and flow batteries are expensive and face scalability issues, making long-duration energy storage a primary research need [73]. Furthermore, grid compatibility remains a concern, as conventional grids were designed for centralized fossil fuel-based generation and require significant upgrades to handle decentralized and variable renewable sources [74]. Several technological innovations hold potential solutions to these challenges. Energy storage innovations, including solid-state batteries, hydrogen-based storage, and pumped hydro, provide pathways to improve energy reliability and address supply fluctuations [73]. The implementation of smart grids, which leverage AI and predictive analytics for real-time demand management and grid optimization, enhances overall system efficiency and reduces curtailment [75]. Additionally, hybrid systems of renewable energy that integrate solar, wind, and other sources in conjunction with energy storage can manage intermittency issues and provide grid resilience [72]. Grid infrastructure enhancement, high-voltage transmission line expansion, and demand-side management strategies also play a crucial role in

ensuring smooth integration of renewable energy in future systems.

3.4. New carbon-efficient feedstocks and sustainable raw materials and production processes

Alternative fuels, feedstocks, and raw materials are critical to lowering carbon emissions in the chemical industry, which is critical for solving climate change problems. Sustainable feedstocks, such as biomass, and CO₂-to-X technologies are developing as viable alternatives to traditional fossil fuel sources. These technologies not only seek to decrease emissions, but also to convert waste carbon into useful products such as methanol and ethanol, so promoting a transition to greener product portfolios [76]. Additionally, sustainable feedstocks including CO₂, biomass, and recycled plastics are crucial for ecologically friendly chemical production. Chemical recycling is a potential means of processing waste plastics and reducing reliance on virgin fossil-based feedstock using pyrolysis, gasification, and solvent-based depolymerization. The use of pyrolysis and other advanced chemical recycling technologies has been shown to mitigate greenhouse gas (GHG) emissions, as shown in recent studies. As an example, a life cycle analysis (LCA) carried out by the University of Warwick indicated that recycling waste plastic using advanced recycling processes is capable of providing a reduction in global warming potential of up to 80 % compared to incineration [77]. Similarly, research in Science of the Total Environment confirmed that pyrolysis chemical recycling is linked to a 50 % lower carbon contribution to climate change and a lower life cycle energy use compared to energy recovery options [78]. Also, chemical recycling increases mechanical recycling by having better material recovery yields and making it possible to produce high-quality feedstock that can be reused to manufacture new chemicals and polymers [79]. These findings underscore the effectiveness of chemical recycling in reducing CO₂ emissions when compared to traditional waste management methods. However, feasibility remains a challenge due to high energy demands, economic viability, and technological scalability.

CO₂, when captured, is transformed into useful products like methanol and polycarbonate polymers. Bio-naphtha, derived from biomass, serves as a renewable resource within the chemical and biofuel sectors, helping to decrease both carbon emissions and fossil fuel dependence through processes such as hydrotreating, pyrolysis, and Fischer-Tropsch synthesis. As a byproduct from producing renewable diesel, bio-naphtha is also used in the petrochemical industry to create eco-friendly products like packaging, fibers, and tires. Moreover, lignocellulosic biomass, sourced from agricultural residues and algae, is processed into biofuels and chemicals, thus minimizing reliance on oil-based feedstocks. Recycling plastics into pyrolysis oil or through mechanical processes also cuts down the need for virgin materials, supporting the principles of a circular economy [80]. Low-carbon fuels such as biofuels, hydrogen, and synthetic natural gas (SNG) offer sustainable alternatives to conventional fossil fuels. Addressing issues like production costs and the need for infrastructure development is crucial. Bioethanol and biodiesel, derived from biomass, serve as replacements for gasoline and diesel, significantly reducing carbon emissions. Green hydrogen, produced by electrolyzing water with renewable energy, only emits water, positioning it as an

exceptionally clean fuel option [81]. Additionally, processes like CO₂ methanation and biogas upgrading generate renewable methane, effectively reducing the carbon footprint of the industry [82, 83]. Alternative raw materials such as lignocellulosic materials and green ammonia provide sustainable sources for chemical manufacturing, with applications ranging from bioethanol production to sustainable fertilizer. These innovative strategies are depicted in Fig. 9, highlighting the variety of alternative fuels, feedstocks, and raw materials significant for chemical companies aiming to reduce their environmental impact.

3.4.1. Methanol production innovations

Innovative methods are being developed to manufacture methanol with minimal CO₂ emissions, which are typically linked with syngas-based approaches. Modern techniques generate syngas by partially oxidizing natural gas or biogas. Technologies such as oxyfuel combustion and gas washing capture and recycle CO₂, reducing greenhouse gas emissions [84]. Using biomass and CO₂ from industrial emissions for methanol production promotes an anthropogenic carbon cycle, resulting in significant emissions reductions. Using collected CO₂ can reduce emissions by half, but relying on finite resources like fossil fuels and biomass may limit sustainability. Integrating bio-methanol and e-methanol processes, which use CO₂ from BECCS/BECCU units and DAC—all powered by renewable energy—could establish a sustainable, carbon-neutral cycle akin to natural photosynthesis. Currently, methanol from non-renewable sources remains integral as a competitive traditional fuel and key industrial feedstock [85]. During the transition to fully renewable sources, hybrid systems could utilize existing infrastructure to support sustainable methanol production, as outlined in the "2023 Sustainability Report," which presents various strategies to reduce carbon emissions and expand the market for low-carbon methanol across three main areas (Fig. 10) [86].

3.4.2. The role of hydrogen

Hydrogen contributes greatly to lowering carbon emissions in the chemical sector through a variety of applications. Hydrogen, as an alternative fuel source, can replace fossil fuels in energy-intensive operations while emitting zero carbon at the point of consumption when produced using renewable energy. It is a major feedstock for making ammonia, methanol, and other chemicals, and using green hydrogen can drastically reduce the carbon footprint of these critical products [11]. Technological advances in hydrogen production are critical to the chemical industry's long-term viability. Green hydrogen, produced by sophisticated electrolysis from renewable sources such as wind, sun, and hydro, emits no carbon. Solid oxide electrolysis cells (SOECs) have increased production efficiency and energy consumption. Blue hydrogen involves Steam Methane Reforming (SMR) with Carbon Capture and Storage (CCS), effectively capturing CO₂ during production, making it a cleaner alternative within industrial applications. Auto-Thermal Reforming (ATR) combines partial oxidation with steam reforming, incorporating Carbon Capture and Storage (CCS) for a more integrated approach. Green hydrogen's use in the Haber-Bosch ammonia synthesis process significantly reduces emissions, with advanced catalysts enhancing efficiency and reducing energy demands. Hydrogen also plays a crucial role as a reducing agent in the sustainable steel

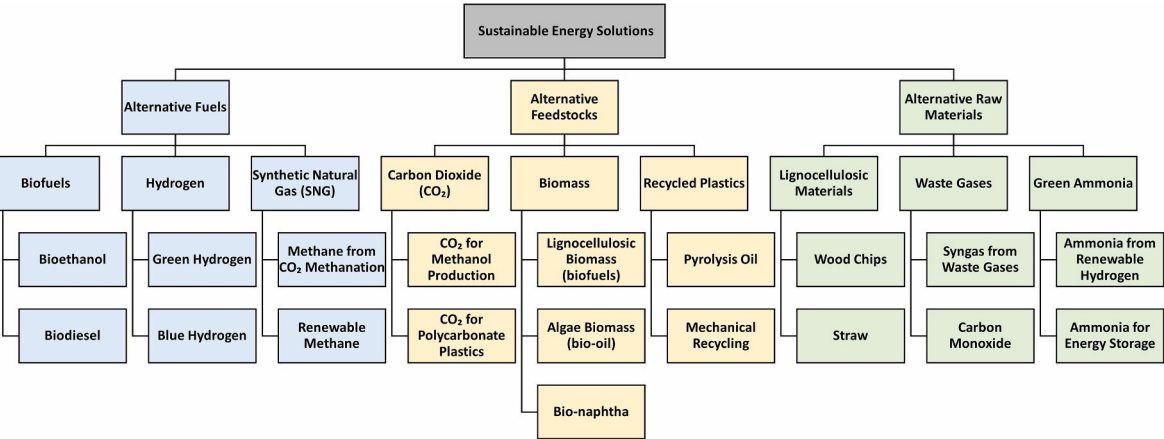


Fig. 9. Alternative fuels, feedstocks, and raw materials in the chemical industry for reducing carbon emissions.

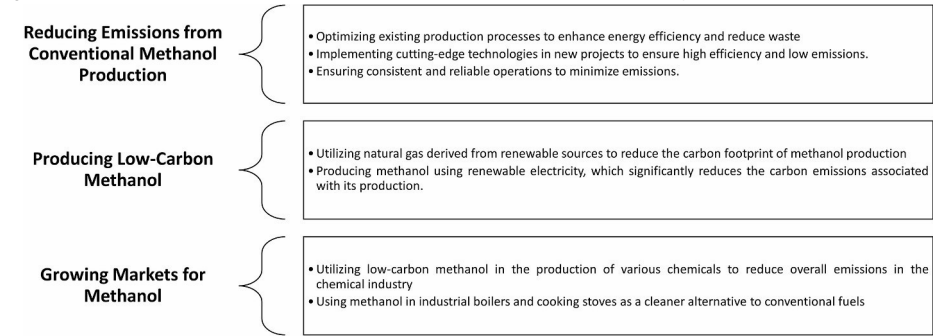


Fig. 10. The strategies for reducing carbon emissions in methanol production.

production via Direct Reduction of Iron (DRI) and supports chemical recycling of plastics. Additionally, hydrogen fuel cells and Combined Heat and Power (CHP) systems provide clean energy solutions, decreasing reliance on fossil fuels. Innovations in hydrogen storage, like solid-state materials and Liquid Organic Hydrogen Carriers (LOHCs), improve the feasibility of hydrogen transport and storage for industrial applications [21,87]. Hydrogen and CO₂-based chemistry are revolutionizing methanol and olefins production to be more environmentally friendly. Traditionally, methanol is produced via coal gasification, especially in Asia, and about 30 % of it goes into producing olefins through the methanol-to-olefins process. An innovative low-carbon method now uses hydrogen and captured CO₂ as feedstocks, utilizing CO₂ from processes like steam methane reforming and emissions from industries such as steel and cement. This alternative approach offers a sustainable route to produce olefins and aromatics without relying on fossil-based petrochemical cracking [20].

3.5. Advanced carbon capture and storage and utilization technologies

industrial sources into valuable products. These technologies are critical for achieving greenhouse gas reduction targets and promoting sustainable industrial practices. Advanced CCU features revolutionary capture techniques, cutting-edge conversion technologies, applicability in a variety of industries, and compatibility with renewable energy sources. Carbon capture, which was previously utilized to improve natural gas quality, now allows for the permanent sequestration of CO₂, as well as the development of techniques to transform waste CO₂ into viable products for industrial and commercial applications Fig. 11

Capture technologies mitigate CO₂ emissions by extracting carbon dioxide from exhaust or processed gases using three main techniques: pre-combustion, post-combustion, and oxyfuel combustion. Both pre-combustion and post-combustion systems use liquid solvents in a packed-bed absorber where CO₂ is captured as gases ascend and the solvent descends, later releasing the CO₂ by altering temperature or pressure to reuse the solvent. This captured CO₂ is then either utilized industrially or sequestered in geological formations or saline aquifers [89]. For pre-combustion, traditional fuels such as coal or natural gas

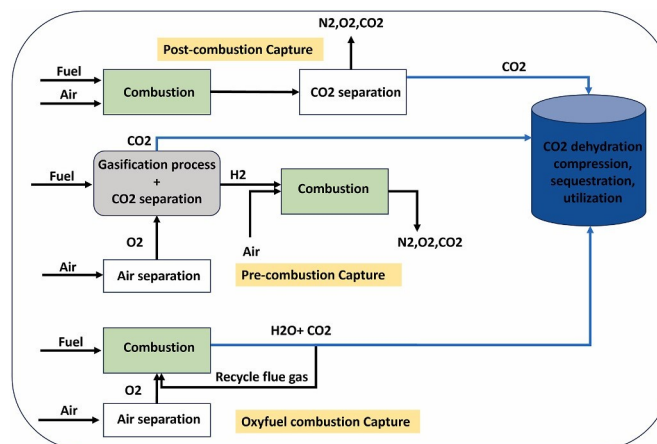
Available CCUS technologies					
Capture (Extract CO ₂ from exhaust or reformed gases)	Sequestration (Store the captured CO ₂ in geological formations)	Enhanced Oil Recovery (Employ methods like CO ₂ injection or thermal recovery to extract additional oil from existing fields)	Utilization (Utilize captured CO ₂ directly, such as in cement production, or convert it into valuable industrial products like chemicals or fuels)	Direct Air Capture (DAC) (Extracts CO ₂ directly from the atmosphere for storage or utilization)	Bio Energy Carbon Capture and Storage (Capture and store CO ₂ from biomass, a renewable energy source)

Fig. 11. Available CCUS-related technologies in chemical industry.

are processed with air or oxygen, often with added steam, to generate synthesis gas, primarily composed of carbon monoxide and hydrogen. This process employs auto-thermal reforming, merging partial oxidation with steam reforming to utilize the heat from oxidation to support the steam reforming process. The resulting syngas is cooled and purified to eliminate contaminants like hydrogen sulphide and mercury. A water-gas shift reaction then increases the CO₂ concentration for more effective separation while also producing hydrogen, which releases only water when combusted. This versatile method, prevalent in the chemical industry, often involves the separation of hydrogen and CO₂ using amine-based solvents through physical or chemical absorption. Despite its utility, pre-combustion requires insubstantial energy for reforming and air separation, driving the development of more efficient technologies like adsorptive and membrane reactors, particularly those using carbon molecular sieve membranes, to enhance the integration of reaction and separation processes [90,91]. Pre-combustion technologies are also becoming recognized for their cost-effectiveness, potentially more so than post-combustion and oxyfuel options, with potential cost savings of 38–45 % and 21–24 %, respectively. However, retrofitting existing facilities with these advanced technologies presents significant logistical and financial challenges [90,92]. Growing interest in post-combustion CO₂ capture is driven by its potential to mitigate climate change and its applicability in enhancing oil and gas recovery or manufacturing chemicals like methanol or dimethylether. Recognized as the most developed and minimally disruptive option for existing setups, post-combustion carbon capture is favored widely. However, its significant energy demands for extracting CO₂ from exhaust gases reduce power plant efficiency by 8–12 %, posing a major economic burden [93]. The typical post-combustion capture process uses MEA (monoethanolamine) absorption without stream division, necessitating substantial steam to regenerate the amine solution [37]. The research [94] has focused on enhancing the cost-efficiency of these systems by optimizing the Heat Exchanger Network of an Absorption Desorption Unit (HEN ADU) using Process Integration techniques and sophisticated computer modeling. Their studies show that integrating compact plate heat exchangers in HEN ADUs can achieve energy savings of up to 13 %, despite similar or lower initial costs compared to traditional heat exchangers. Oxy-fuel combustion technology transforms traditional combustion by utilizing oxygen of over 95 % purity instead of air and incorporating recirculated flue gases [93]. This approach results in a flue gas primarily composed of carbon dioxide and water vapor, the latter of which is easily condensed and removed. The resulting purified CO₂ can be compressed and either stored in geological formations or used in various applications such as geothermal energy production, agriculture, or the synthesis of biofuels like methanol and dimethyl ether, forming an integral part of a CCUS system [88]. While economically challenging due to the costs associated with oxygen production and separation, oxy-fuel combustion simplifies the CO₂ capture and separation process, incurring only a 4 % efficiency loss compared to the 8–12 % typical of post-combustion techniques [93]. The use of oxy-fuel combustion in power plants reduces the volume of exhaust gases and nitrogen oxide emissions, enhances boiler efficiency, and stabilizes temperatures, but the requirement for high-purity oxygen remains a costly and resource-intensive barrier [90]. Despite these challenges, the benefits, including reduced flue gas volume, higher boiler efficiency, elimination of nitrogen oxides, and potential for retrofitting existing facilities, are significant [95]. A schematic of the three carbon capture processes is

illustrated in Fig. 12. This figure provides a visual representation of each method, highlighting the key components and stages involved in capturing CO₂ from industrial processes.

Numerous studies within the chemical industry focus on evaluating



diverse carbon capture methods to enhance the reduction of greenhouse gas emissions. Gabrielli et al. explore three strategies to reach carbon neutrality in this sector: employing fossil fuels combined with CCS, utilizing green hydrogen with CCU, and leveraging biomass (BIO). Each method presents distinct advantages and inherent challenges: CCS capitalizes on pre-existing infrastructure but depends on fossil fuels and CO₂ storage; CCU necessitates considerable electricity from renewable sources; and BIO requires substantial land use, which could lead to conflicts. Thus, comprehensive system analysis and life cycle assessments (LCAs) are essential to gauge the effects of these approaches on electricity use, land deployment, and CO₂ emissions [3].

Expanding on their fundamental research, Katelhⁿ on et al. explore the " potential of carbon capture and utilization (CCU) to mitigate climate change impacts within the chemical sector, presenting three distinct scenarios aimed at achieving carbon neutrality by 2030. The traditional scenario projects increased emissions due to rising demand. Conversely, a high TRL scenario employs CO₂-derived methanol and methane to significantly boost feedstock mass flows, making use of existing infrastructure. In contrast, the low-TRL scenario focuses on direct CO₂ conversion into chemicals, which requires less CO₂ and hydrogen but relies on nascent technologies and extensive research and development. Both CCU scenarios highlight a substantial increase in electricity demand, emphasizing the critical need for significant expansion in low-carbon electricity generation to effectively reduce greenhouse gas emissions [96]. Further research [97] examines the environmental impacts of post-combustion carbon capture at natural gas power plants in Iraq, particularly the trade-offs involved in retrofitting plants versus building new ones. Their findings highlight a reduction in global warming potential offset by increases in acidification and particulate matter, pointing to the complexities of environmental impacts in carbon capture decisions. Additionally, Myrian Schenk and Jim Middleton from Technip Energies explore the integration of carbon capture in ethylene production facilities. Their research examines strategies such as fuel substitution and electric heating, combined with carbon capture, to achieve significant emission reductions in ethylene manufacturing [98].

3.5.1. Enhancing carbon capture efficiency

Post-combustion carbon capture uses amine solvents to remove CO₂ from flue gases after combustion. The CO₂ binds with

solvents like MEA, diethanolamine (DEA), or methyldiethanolamine (MDEA) in an absorber unit, and is later released through heating in a regeneration column. The captured CO₂ is then compressed for geological storage or reused in industrial applications, such as enhanced oil recovery or chemical production [3]. Traditional CO₂ capture with MEA is

Fig. 12. Carbon capture processes.

sumption, and lowering operational costs, making carbon capture more viable [51,100]. Research [101] on high concentration triethylenetetramine (TETA) in organic solvents shows improved CO₂ absorption rates and lower energy needs during regeneration. Microwave-assisted regeneration further enhances energy savings compared to traditional methods. These nonaqueous absorbents can also effectively manage water vapor in flue gases, making them well-suited for industrial applications. McGurk et al. demonstrated that microwave irradiation can regenerate MEA solutions at lower temperatures (70–90 °C), reducing operational costs. Their research emphasizes the 'non-thermal' effects of microwaves, enabling more efficient CO₂ desorption compared to traditional methods [99]. Beyond these improvements in amine-based capture, DAC represents a significant advancement in capturing CO₂ directly from the atmosphere. Captured CO₂ can be stored or used to produce synthetic fuels, chemicals, or for enhanced oil recovery. Combining CCS with DAC offers a comprehensive solution to reduce atmospheric CO₂ levels, with CCS addressing emissions from industrial sources and DAC targeting existing atmospheric CO₂. This integrated

approach is crucial for global carbon reduction and climate change mitigation efforts. Biochar is increasingly recognized as a promising negative emissions technology. Industrial-scale biochar systems have demonstrated the ability to remove approximately 2.68 tCO₂e per tonne of biochar produced, with a total sequestration potential of 24,450 tCO₂e annually when widely implemented [102]. The viability of integrating biochar into carbon reduction strategies depends on key factors such as feedstock availability, pyrolysis efficiency, and market incentives, particularly for its application in soil enhancement and carbon credit markets.

3.5.2. CO₂ Utilization Pathways

A promising strategy to reduce the chemical sector's carbon footprint is using CO₂ as a feedstock for bulk product production, helping to lower emissions. However, CO₂-based manufacturing is less cost-effective than traditional petrochemical methods and requires building a new value chain. With CCS storage capacity limitations, it is essential to develop sustainable CO₂ utilization methods. These include innovative and sustainable techniques such as fermentation and algae cultivation for biofuel production, hydrogenation to produce dimethyl ether (DME), methanol conversion, and the synthesis of tertiary butyl methyl ether (MTBE). CO₂ also serves as a feedstock for urea production, polycarbonate plastics, and salicylic acid. Additionally, mineralization combines CO₂ with minerals to create stable carbonates for construction materials or concrete, enhancing properties while sequestering CO₂. Enhanced oil recovery (EOR) is another vital application, involving injecting CO₂ into oil fields to increase crude oil extraction while storing CO₂ underground. CO₂ is also utilized in biological processes, including microbial conversion and algae cultivation, to create biofuels, bioplastics, and animal feed. Advanced chemical synthesis pathways include electrochemical reduction, forming various chemicals like formic acid, carbon monoxide, and ethylene, and photocatalytic conversion, where CO₂ is transformed into fuels and chemicals using sunlight. The electrochemical reduction of CO₂ is emerging as a highly promising method to mitigate the impact of increased atmospheric CO₂ concentrations, effectively converting CO₂ into valuable chemicals like methanol, ethanol, and formate [103]. Photocatalysis, ranging from traditional semiconductor-based approaches to novel piezo-electrocatalysis and hybrid systems, provides exciting possibilities for efficiently and sustainably converting CO₂ into valuable goods. These approaches utilize materials such as dual-metal-loaded metal oxides or carbon-doped TiO₂ to absorb solar energy and transform CO₂ into useful chemicals or fuels [104]. Economic closed-loop CO₂ recycling systems, tailored for on-site implementation, are poised for

energy-intensive, requiring steam heating at 120–140 °C under 1–2 bar pressure, which increases costs and causes amine degradation [99]. Microwave heating offers a promising alternative by selectively heating the amine-CO₂ complex, improving efficiency, reducing energy con-

quick adoption across industries. These retrofit-driven recycling initiatives could slash annual CO₂ emissions by 4–10 Gt by 2050, potentially accounting for up to 50 % of the industry's goals for carbon neutrality [105]. The study [106] explores the potential of Sn-based electrocatalysts in the electrochemical CO₂ reduction reaction (eCO₂RR) to produce formic acid, noting their high revenue potential and low toxicity. Despite their high selectivity, these catalysts face challenges with stability, requiring extensive operational hours to be economically viable. The research sheds light on the degradation mechanisms impacting long-term activity and underscores the need to optimize reactor designs and process conditions for stability. Further, converting CO₂ into valuable compounds such as methane, methanol, or more complex alcohols through power-to-gas technology is crucial for reducing emissions. Methanation, for instance, transforms CO₂ into synthetic natural gas (SNG), producing methane, while other processes convert CO₂ to carbon monoxide for Fischer-Tropsch synthesis, creating liquid hydrocarbons like diesel and kerosene [107]. Yusak et al. demonstrated CO₂ methanation using nickel nanoparticle-doped mesoporous carbon nitride catalysts, achieving high methane selectivity and showing strong potential for reducing CO₂ emissions and mitigating global warming [108]. Massa et al. explored sorption-enhanced methanation (SEM), which uses in-situ steam removal to boost methane conversion efficiency. While effective, they identified challenges like carbon formation at high temperatures and low pressures, highlighting the need for controlled steam capture to prevent carbon buildup [109]. Moreover, CO₂ is being utilized to produce organic carbonates like dimethyl carbonate (DMC), oxalic acid through oxidative coupling, and propylene carbonate via cycloaddition reactions. Enhanced Geothermal Systems (EGS) and baking soda production are also notable applications of CO₂, using it as a working fluid and a reactant, respectively. Each pathway offers distinct benefits and challenges, with ongoing research aimed at optimizing these processes to boost CO₂ utilization efficiency and economic viability. This concerted effort underscores the industry's commitment to innovative solutions for environmental sustainability. A summary of the pathways for CO₂ utilization in the chemical industry discussed in the previous paragraphs is presented in Fig. 13.

3.6. Innovative carbon reduction

In addressing climate change, chemical companies are vigorously adopting measures to reduce CO₂ emissions through a variety of strategies, including renewable energy adoption, novel sustainable production technologies, and process optimizations. To provide a comprehensive overview, Table 1 presents specific case studies of chemical companies, detailing their implemented strategies, achieved reductions, and encountered challenges.

This structured summary allows for an in-depth understanding of the industry's efforts while ensuring clarity and conciseness.

The collective efforts of companies in the chemical sector signify a major shift toward sustainable technologies and processes, not just to lessen environmental impacts but to also set new industry standards in ecological responsibility. These initiatives demonstrate a strong commitment to integrating advanced technologies to significantly cut carbon emissions, thereby aligning with global efforts to mitigate climate change.

4. Challenges and barriers to achieving carbon neutrality in the chemical industry

Energy-intensive industries, such as chemicals and steel, face significant challenges in transitioning to clean energy technologies while maintaining competitiveness. In these sectors, where profit margins are slim, the high upfront costs of technologies still in the demonstration phase, such as electric motors and heat pumps, present major hurdles. International competition adds pressure, making industries cautious about absorbing higher production costs, leading to the implementation of carbon border adjustment measures to maintain competitiveness while complying with environmental regulations [11]. The 40-year lifespan of heavy industry assets, such as blast furnaces and cement kilns, highlights the urgency of transitioning to cleaner technologies to meet the 2050 carbon neutrality deadline. The next decade is critical for adopting near-zero emissions technologies, which are still in the prototype or demonstration phase, to ensure timely progress toward decarbonization. With 30 % of existing assets soon needing major refurbishments or replacements, this presents a key opportunity to adopt new technologies. Failure to act could hinder emission reduction efforts and result in stranded assets, jeopardizing progress toward carbon neutrality [32]. The transition to renewable energy presents not only technical and economic challenges but also significant social implications that must be addressed to ensure an inclusive and just energy transition. As highlighted in recent literature [118], the social, environmental, and economic consequences of renewable energy integration extend beyond emissions reduction, influencing public acceptance, employment structures, and

renewable energy are not evenly distributed, with urban areas gaining economic advantages while rural and marginalized communities face higher costs and land-use conflicts. There is a need for targeted policies such as subsidies and labor force upgradation to enable a fair transition. Since achieving significant reductions in carbon intensity relies on CCUS technology, addressing its limitations is essential for large-scale adoption and progress toward carbon neutrality. One of CCUS technology's primary limitations is high energy demands in CO₂ capture and regeneration processes, particularly in post-combustion and direct air capture systems. These energy requirements often offset a portion of the environmental benefits unless integrated with low-carbon or renewable energy sources. Additionally, CCUS requires substantial infrastructure, including pipelines for CO₂ transport and storage facilities, which pose significant logistical and financial challenges, especially for geographically dispersed industries [119]. Economic feasibility is one of the key challenges to large-scale deployment of CCUS technologies. The cost of using CCUS—ranging from capital expenses to operational expenses, to transportation and storage of CO₂—is generally too high without government subsidies or robust carbon pricing mechanisms. Overcoming these economic challenges requires advancements in technology, economies of scale, and supportive policy frameworks to reduce costs and incentivize adoption. Despite its challenges, CCUS hold significant potential for reducing emissions in sectors that are challenging to decarbonize, such as chemicals, cement, and steel. Advances in CO₂ utilization technologies, including low-energy catalytic processes and renewable energy integration, present opportunities to mitigate the energy penalties traditionally associated with CCUS. Moreover, CCUS offers a dual benefit: it not only reduces emissions but also enables the production of valuable by-products such as synthetic fuels and carbon-based materials, which can generate additional revenue streams to offset implementation costs. As highlighted by Hepburn et al. [120], the success of CCUS technologies will depend on their scalability, cost reductions, and the development of markets for CO₂-derived products. With these technologies coupled with enabling policies, CCUS would be a transformative option for carbon neutrality in sectors that generate unavoidable process emissions. Detailed studies, such

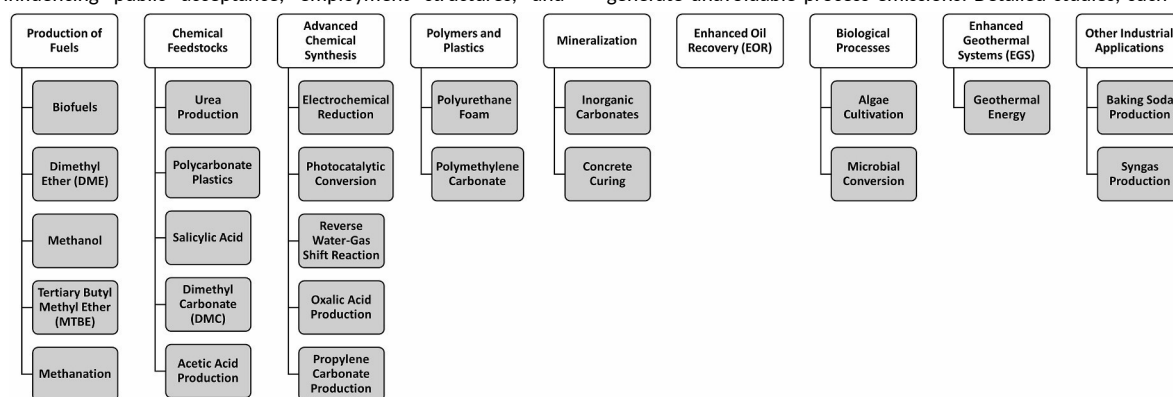


Fig. 13. Classification of CO₂ utilization pathways in the chemical industry.

socioeconomic equity. Public acceptance remains a challenge for large-scale renewable projects due to concerns over land use, environmental impacts, and community disruptions. Transparent policies and stakeholder engagement are crucial for gaining support. The transition to renewable energy is also reshaping employment, creating new employment in solar, wind, and energy storage industries, but threatening existing employment in fossil fuels. Reskilling programs become crucial to enable adjustment of the workforce. Additionally, the benefits of

as those by Gabrielli et al. [3], explore various technological pathways toward a carbon-neutral chemical industry. However, each option presents challenges: biomass cultivation requires large areas of land, and CCU has high energy demands. The feasibility of these technologies often depends on external factors, such as CO₂ storage availability and the carbon intensity of the power used. Scaling these solutions will require extensive CO₂ transport and storage infrastructure, adding complexity to planning and development.

Furthermore, the transition to sustainable systems in sectors like aviation fuel and chemical production is constrained by the limited availability of essential resources, such as sustainable biomass. Adopting emerging technologies like microwave technology in carbon capture faces challenges with scalability, integration, and achieving uniform heating. Similarly, CO₂ electrochemical reduction encounters obstacles related to catalyst stability and by-product purity, making industrial scalability difficult. As industries aim to transition to greener practices, they face hurdles such as the high costs of renewable technologies like methanol production, which require substantial financial and policy support. The integration of hydrogen and hybrid heat supply systems to meet rising electricity demand further emphasizes the need for cross-sector collaboration and infrastructure development [121]. The effectiveness of GHG emission reductions through CCU depends heavily on the carbon intensity of the electricity used. Substantial reductions can only be achieved with access to low-carbon or carbon-free power sources, especially for processes like water electrolysis in the CCU-DAC framework, which requires a fully carbon-neutral electricity supply [96]. A significant challenge in adopting sustainable technologies like BECCS in aviation and chemical sectors is the limited availability of sustainable biomass, which is critical for scaling these negative emission technologies. This restricted supply could hinder broader adoption. To achieve large-scale net-zero CO₂ emissions, a comprehensive CO₂ network is essential, capable of transporting CO₂ from various sources to

multiple storage and utilization sites. Developing this infrastructure effectively during the transition phase presents substantial challenges due to existing inadequacies. Accurate CO₂ footprint assessment requires shifting from cradle-to-gate to cradle-to-grave analyses, considering the full lifecycle of products, including disposal, to capture all related emissions [3]. Nickel-based catalysts, supported by alumina, have shown high efficacy in CO₂ methanation due to their selectivity and activity, making them suitable for large-scale applications. However, their widespread adoption is limited by the scarcity and cost of noble metals and their susceptibility to deactivation, which challenges their efficiency and long-term sustainability [108,122]. CO₂ electrochemical reduction also faces challenges, such as catalyst stability, selectivity, competing reactions, and impurity management, which impede its scalability and industrial application [106]. The adoption of renewable methanol is hindered by its higher production costs compared to traditional methods. Policy support, enhanced carbon pricing, and subsidies are essential to facilitate the transition to bio-methanol and e-methanol, which require substantial investment and stable regulatory frameworks [83]. Challenges in olefin production include high costs, technological immaturity, and significant infrastructural and policy support needs, all of which are vital for integrating olefin production into a sustainable, circular economy [123]. As industries move towards process electrification and hydrogen production, the increasing demand for electricity necessitates a reduction in reliance

Company	Carbon reduction measures	Remarks	Company	Carbon reduction measures	Remarks with environmental sustainability goals.
BASF [110]	- Increasing renewable energy use - Implementing power-to- steam via electric heat pumps to generate CO₂- neutral steam from waste heat. - Deploying electricity-based technologies including e- Boilers, e-furnaces, e-Drives, and heat pumps - Producing hydrogen without CO₂ emissions - Achieving CO₂-free methanol production - Utilizing renewable raw materials - Optimizing energy production and manufacturing processes - Reducing emissions throughout the value chain	- Aiming for net-zero emissions by 2050. - Investing in renewable energy production and infrastructure to cut CO₂ emissions by at least 90 % in key chemical processes. - BASF, SABIC, and Linde are developing the world's first large-scale electric steam cracker furnace demonstration plant. - Exploring clean hydrogen production through innovative methane pyrolysis and established water electrolysis. - Developed an environmentally friendly methanol production method using syngas from partial oxidation of natural gas or biogas, incorporating oxyfuel combustion, OASE® gas washing, and CO₂-free hydrogen. - BASF promotes energy efficiency with its Energy Verbund and combined heat and power plants, saving 19.0 million MWh and reducing CO₂ emissions by 6.2 million metric tons in 2022 by optimizing energy use and recycling waste heat. - Committed to addressing climate change, supporting a circular economy, and conserving biodiversity. - Aiming for a 30 % reduction in its carbon footprint by 2030, in line with Science Based Target initiatives (SBTi). - Launching the UK's first large-scale CCU facility. - Significant progress noted with the UK's CCU plant and the expanded use of natural gas in	Dow [112, 113]	- E-cracking technologies for efficient ethylene production - Advanced Propane Dehydrogenation technology for improved yield - Techniques for converting cracker off-gas into hydrogen and capturing CO₂ for storage or utilization	- Dow and Shell are partnering to develop electric cracking technologies that utilize renewable energy, aiming to significantly reduce emissions. - At its Plaquemine complex in Louisiana, Dow has implemented propane dehydrogenation technology, paving the way for future ethane dehydrogenation methods and moving away from traditional high-emission ethylene production. - Dow is planning to build the world's first net-zero carbon ethylene and derivatives complex, which will decarbonize 20 % of its global ethylene capacity, increase polyethylene supply by 15 %, and substantially boost EBITDA. - LG Chemical in Korea aims for "Carbon-neutral Growth" by 2030 and "Net-Zero Emissions" by 2050. - Plans to switch to 100 % renewable energy for overseas facilities by 2030 and domestic facilities by 2050. - Expanding recycling of waste plastics and batteries, promoting a Zero Waste to Landfill policy across all operations. - Minimizing the use of fossil fuels in naphtha cracking furnaces and converting coal boilers to biomass at the Yeosu Hwachi Complex. - Initiating the petrochemical industry's first hydrogen production facility using
Tata Chemicals [111]	- Reducing, reusing, and recycling natural resources and waste. - Switching to low-carbon fuels to decrease emissions. - Increasing the use of renewable power sources. - Enhancing overall energy efficiency. - Implementing CCU technologies.		LG Chemical [114,115]	- Adoption of eco-friendly fuels to minimize environmental impact. - Advancements in Carbon Capture and Utilization (CCU) technology to improve sustainability. - Enhancements in energy efficiency across industrial operations. - Development of mechanical and chemical recycling methods, and production of biodegradable plastics. - Implementation of electrolytic furnaces for cleaner production processes.	

Finnish startup
Coolbrook
[85]

- Electrify the cracking process with the Rotor Dynamic Reactor

US operations, while the Mithapur manufacturing complex has achieved freshwater neutrality.

- At Mithapur, initiatives like MUW FBD and vacuum system modifications, coupled with increased renewable energy generation and bicarbonate production, have significantly reduced emissions.
- This innovative technique utilizes a turbomechanical strategy to significantly enhance ethylene production, increasing yields by up to 30 % while simultaneously reducing energy consumption by the same margin. When powered by renewable energy sources, it enables emissions-free operations, aligning

Table 1 Table 1 (continued) measures to reduce CO2 emissions by chemical companies.

Mitsui
Chemicals
[116]

- Adopting low-carbon materials and fuels
 - Enhancing energy efficiency across operations
 - Transitioning to renewable energy sources
 - Developing innovative processing technologies
 - Implementing a highly efficient gas turbine power generation system for energy self-sufficiency
 - Utilizing LNG cold energy for energy-saving processes
 - Employing Carbon Capture Utilization (CCU) technologies
- methane from Naphtha Cracking Center to reduce emissions.
- Collaborating with Taekyung Chemical to convert CO2 byproduct from hydrogen production into dry ice.
 - Committed to achieving carbon neutrality by 2050, with a specific target set in June 2021 to reduce global GHG emissions by 40 % from 2013 levels by 2030.
 - Reducing annual CO2 emissions from Osaka Works by 70,000 tons through the installation of a Highly Efficient Gas Turbine.

(continued on next page)

Table 1 (continued)

Company	Carbon reduction measures	Remarks
Cosmo Energy Group [117]	• Switching naphtha crackers to alternative fuels, such as ammonia • Move to Decarbonized Energy • Implementation of Negative Emissions Technologies (such as CCS/CCUS, CO₂-EOR, etc.) • Strategies for Advanced Energy and Raw Material Solutions	• The Group is actively reducing Scope 1 and Scope 2 emissions by adopting lower-emission fuels like LNG, hydrogen, and ammonia, integrating renewable energy, and improving energy conservation. This initiative is part of a larger effort to transition all electricity used at Cosmo Oil Sales' service stations to renewable sources. • The Group is assessing the feasibility of using CO₂-EOR, CCS, and CCUS technologies in its operations and has partnered with the Abu Dhabi National Oil Company to explore decarbonization technologies and conduct feasibility studies on CCS/CCUS in Abu Dhabi. • Leveraging its hydrogen expertise, the Group is considering expanding into the hydrogen station business and developing a global hydrogen supply chain. It has also partnered with UAE-based Masdar to explore collaborations in decarbonization projects, including wind farms and hydrogen and ammonia projects. • Creating a bio-jet fuel supply chain from used cooking oil has been selected by NEDO, aiming to start full-scale sustainable aviation fuel supply by 2025.

on fossil fuels. Integrating hydrogen and hybrid heat supply systems enhances flexibility, allowing the chemical industry to optimize costs by aligning energy purchases with low-price periods. This transformation will necessitate cross-sector collaboration to create the infrastructure required for managing power, hydrogen, and CO₂ [124]. While bio-based materials reduce carbon footprints, large-scale adoption raises concerns about deforestation, agricultural displacement, and water scarcity [125,126]. Intensive biomass cultivation can also increase fertilizer and pesticide use, contributing to soil degradation and water pollution [127]. To mitigate these impacts, waste biomass, non-food feedstocks, and algae-based biofuels offer promising solutions with lower environmental costs. Circular economy approaches, such as biorefinery integration and bio-based material recycling, further enhance sustainability [128]. A balanced approach combining policy measures, sustainable agriculture, and technological advancements is essential to maximize benefits while minimizing trade-offs.

5. Environmental and economic impacts

Efforts to achieve carbon-neutral growth in the chemical industry focus on adopting eco-friendly fuels, using low-carbon raw materials, advancing Carbon Capture and Utilization (CCU) technology, and enhancing energy efficiency. The adoption of a circular economy model, including advances in mechanical and chemical recycling and the production of biodegradable plastics, plays a vital role in sustainability. Supporting environmental policies such as Zero Waste to Landfill is crucial for resource efficiency and

environmental protection. Additionally, successfully transitioning to carbon neutrality, which balances emitted carbon by sequestering or offsetting an equivalent amount to achieve a net-zero carbon footprint, involves understanding market demands, adapting commercial models, and managing decarbonization as a transformative process [15]. This approach not only addresses environmental challenges but also supports economic and societal benefits by promoting sustainable practices and green technologies, enhancing global climate change mitigation efforts. The integration of these strategies is key to advancing the sustainability goals of the chemical industry.

5.1. Reduction in carbon emissions and environmental benefits

Carbon neutrality offers deep environmental benefits related to sustaining ecosystems and human health on our planet. These benefits include mitigating climate change, preserving biodiversity, and enhancing air and water quality-all of which are essential elements in making the Earth resilient and livable. Carbon neutrality helps reverse environmental degradation and paves the way for a sustainable future for upcoming generations. One of its most immediate and vital impacts is the reduction of environmental degradation and the deceleration of global temperature increases [129]. Integrating LCA data provides a quantitative perspective on the environmental benefits of various decarbonization strategies in the chemical industry. For instance, the electrification of steam cracking processes via renewable power has been shown to reduce CO₂ emissions significantly. A study by Amghizar et al. [130] indicates that such electrification can lead to substantial decreases in greenhouse gas emissions, highlighting its potential as a scalable decarbonization strategy. Additionally, the adoption of circular economy approaches, including the use of bio-based feedstocks, can further contribute to emission reductions [131]. discuss the environmental advantages of bio-based products, emphasizing the importance of sustainable sourcing practices to maximize CO₂ emission reductions. A LCA-based study comparing a base case with a heat recovery scenario for capturing 300 kilotonnes of CO₂ annually from natural gas combined cycle (NGCC) flue gas demonstrated that the heat recovery scenario outperformed the base case in all environmental impact categories. This indicates substantial environmental benefits associated with advanced heat recovery systems [132]. These insights underscore the critical role of innovative technologies and sustainable practices in achieving carbon neutrality within the chemical sector. Carbon neutrality helps to preserve vulnerable ecosystems, such as coral reefs and polar regions, which are sensitive to temperature changes. Reducing reliance on fossil fuels, which are important causes of air and water pollution such as sulfur dioxide, nitrogen oxides, and particulate matter, is a critical tactic in this effort. The transition to cleaner energy sources dramatically reduces emissions of these toxic pollutants, hence improving air and water quality. Additionally, achieving carbon neutrality is critical for avoiding severe climate change consequences such as extreme weather, rising sea levels, and biodiversity loss, hence safeguarding the planet's long-term climate and environmental stability [133].

5.2. Economic implications and profitability of carbon neutrality efforts

Achieving carbon neutrality is poised to fundamentally transform economic development and energy production and consumption, ushering in a new era based on sustainable energy practices. This transition is set to drive significant technological innovations in decarbonization, energy efficiency, recycling, advanced power systems, and negative emissions technologies [5]. The move toward carbon neutrality will reshape the energy sector, particularly impacting high-emission industries such as oil, coal, and natural gas. The oil industry may experience a gradual decline in demand for refined products, while countries dependent on crude oil exports are likely to see demand stabilize and eventually decline as they pivot towards more sustainable energy sources. Similarly, the coal industry will face pressures to reduce production capacities, and natural gas will likely see reduced demand as cleaner energy alternatives become more viable [134]. This shift will accelerate the development of non-fossil fuel energy technologies, essential

for crafting a new energy economy that supports global carbon reduction targets and climate change mitigation efforts. The transition to carbon neutrality represents not just an environmental imperative but also a catalyst for broad economic transformation. It underscores the critical role industries will play in fostering a renewable and efficient economy, enhancing both ecological and economic sustainability and equipping industries for a thriving, low-carbon future. The transition to carbon-neutral operations in the chemical industry involves significant capital investments and operational costs, with financial feasibility being a key determinant of implementation success. Carbon-reduction methods like CCUS, use of renewable energy, and process optimization provide sustainability advantages and financial economies in the long-term, though their initial costs and financial viability differ greatly based on support provided by regulation and technical readiness levels [135]. The cost of adopting renewable energy, such as solar and wind power, has significantly decreased; since 2010, the leveled costs of electricity (LCOE) for solar PV have dropped by more than 80 % [136]. However, high-temperature process industries still struggle in their direct electrification in that they must resort to substitute energy alternatives such as green hydrogen, which is currently 2–4 times more expensive in comparison to conventional fossil fuels [137]. The cost-effectiveness of CCUS technologies is a primary challenge, with capture prices ranging from \$50 to \$200 per ton of avoided CO₂ depending on the capture method and scale of deployment [135]. The construction of infrastructure to transport CO₂ over distances and to store it in the long term also presents financial challenges that require high policy support and carbon pricing to make it more economically viable. In the case of green chemistry and circular economy approaches, production costs for bio-based alternatives often exceed those of fossil-derived chemicals due to higher feedstock costs, process inefficiencies, and scalability constraints [128]. However, with advancements in biorefining technologies and regulatory incentives such as carbon taxes and subsidies, economic competitiveness is improving. According to a lifecycle cost analysis, bio-based polymers are currently 30–50 % more expensive than conventional plastics, but their lower carbon footprint and long-term sustainability advantages make them worth the investment [138]. Overcoming the financial constraints to deploying renewable energy technology in the chemical sector

enhancing long-term cost-effectiveness. The appropriate finance strategy will be linked to the technological maturity, local policy climate, and economic context of the deployment. While no single method is universally desirable, a hybrid strategy that incorporates these approaches can minimize risk, enhance financing availability, and accelerate the commercial viability of industrial decarbonization technologies [142]. Additionally, achieving carbon neutrality in the chemical industry involves navigating a complex landscape of costs and benefits that affect both immediate financial results and long-term profitability, as elaborated in Fig. 14.

This figure presents a quick overview of the financial implications of achieving carbon neutrality in the chemical industry, classifying the data into costs and benefits. Under "Costs," it comprises Capital Expenditure for upgrading to energy-efficient technologies, Operational Changes as a result of using greener materials, and Regulatory Compliance expenditures. The "Benefits" section highlights gain such as lower energy costs, access to incentives, increased competitiveness, and avoidance of regulatory risk. Strategic considerations are also discussed, highlighting the significance of balancing expenses and long-term savings, embracing innovation, and engaging stakeholders in order to effectively manage the shift. This classification helps to understand the economic drivers moving the sector toward sustainability.

With regard to the social implications of achieving carbon neutrality, not only in the chemical industry but also at the global level, there is strong evidence suggesting that a shift toward renewables and other low carbon initiatives improves social outcomes. These outcomes include engagement in the energy system through enhanced energy access, net-positive employment outcomes, improvement in health through the reduction of airborne pollutants and the alleviation of energy poverty in both developed and developing nations [143].

6. Discussion

6.1. Summary of key findings on new and emerging carbon reduction strategies for chemical companies

The chemical industry is a significant contributor to global CO₂ emissions due to its high energy demands and reliance on carbon-based

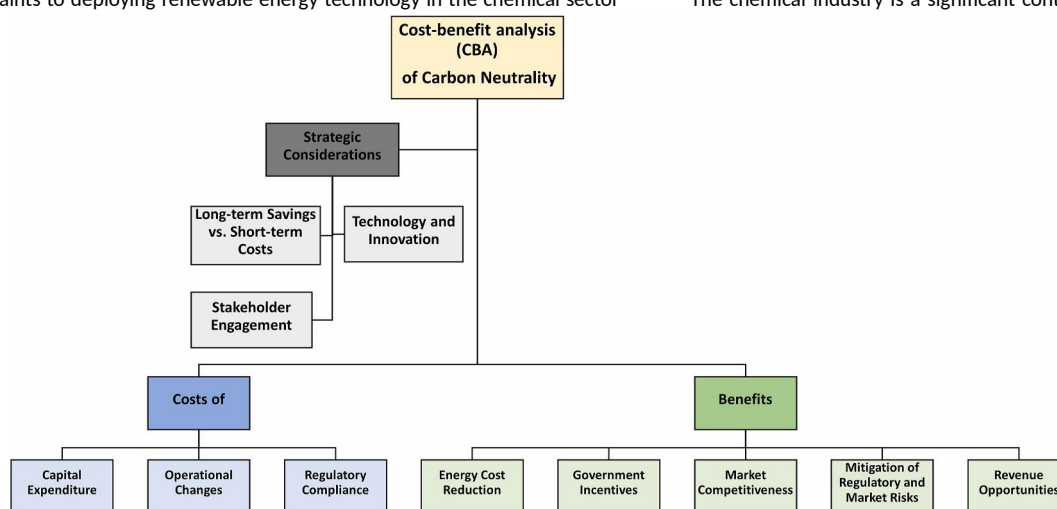


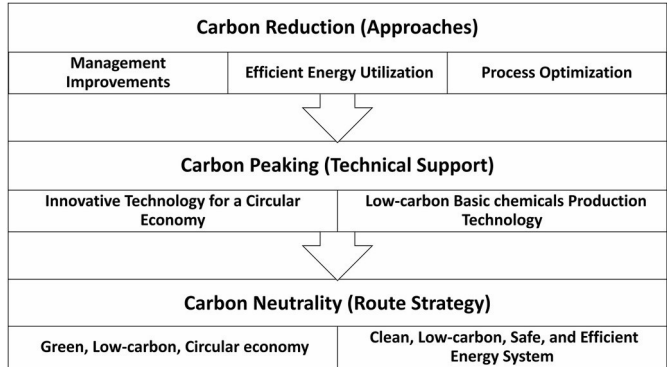
Fig. 14. Cost-benefit analysis for carbon reduction solutions.

would require a combination of technological innovation and supportive governmental tools. Although they frequently rely on ongoing political will and financial resources, capital subsidies and tax incentives can lower the upfront costs of investments [139]. Carbon pricing and emissions trading schemes (ETS) integrate environmental costs while providing market-based incentives, but they are prone to price volatility and regulatory uncertainty [140]. Public-private partnerships (PPPs) and green bonds provide alternative financing structures that spread risk and promote innovation, but they necessitate strong governance and coordination [141]. Efforts to reduce lifecycle costs through economies of scale, innovation, and integration with CO₂ usage pathways, such as synthetic fuels or materials, are crucial for

feedstocks. Achieving carbon neutrality necessitates a multifaceted approach that integrates technological advancements, process optimizations, policy frameworks, and circular economy models. As global climate policies aim for peak emissions by 2030 and net-zero emissions by 2050, the chemical sector must overcome challenges associated with both energy-related and process-based emissions [121,144]. This section outlines the strategic roadmap (Fig. 15) for transitioning the chemical industry toward carbon neutrality through a structured approach that includes emission reduction, circular economy integration, and long-term sustainability strategies.

The first phase is centered on minimizing emissions at the source through industrial process optimization and energy efficiency improvements.

Technologies such as heat integration and waste heat recovery via ORC and high-efficiency heat exchangers can greatly reduce energy use and CO₂ emissions. Additionally, digitalization and AI-driven energy management enhance operational efficiency by predicting maintenance needs and optimizing resource utilization. Electrification of high-temperature processes such as steam cracking is also a likely path to emission reductions, particularly when it is done using renewable energy sources. To further reduce reliance on fossil fuels, chemical industries can employ renewable energy, produce ammonia and methanol using green hydrogen, and employ hybrid heat supply systems to enable low-cost and flexible energy management. Beyond emission reductions, carbon peaking involves the integration of advanced technologies and circular economy principles. CCUS technologies have demonstrated up to 90 % CO₂ capture efficiency, but their economic feasibility and infrastructure development remain key challenges for widespread adoption. Within the circular economy framework, chemical recycling methods offer a sustainable alternative to conventional waste management, reducing emissions compared to incineration and landfilling. Additionally, shifting from fossil-based feedstocks to bio- based alternatives has the potential to lower carbon intensity, provided that sustainable sourcing and LCA are considered. Hydrogen- based processes, particularly for CO₂-to-methanol and olefins production, offer a viable alternative to traditional petrochemical processes, further supporting decarbonization efforts. For long-term carbon neutrality, grid infrastructure optimization and large-scale adoption of renewable energy are crucial. Smart grids and AI-based energy forecasting can support better distribution of energy and efficiency, making it possible to transition to fully electrified processes in industry. Large- scale carbon capture and negative emission technologies provide more



possibilities for carbon use and storage in the long term. Enhanced CO₂ utilization technologies also close the carbon loop and facilitate circularity in manufacturing processes.

Fig. 15. Strategy diagram of the necessary phases and methods to achieve carbon neutrality in the chemical industry.

enhance progress, global agreements focused on key sectors like chemicals are essential for fostering international collaboration and technological exchange.

Support Infrastructure Development: Investing in CO₂ transport and storage infrastructure is essential to support CCS technologies and integrate them into existing workflows [154]. Additionally, building infrastructure for low-emission hydrogen production and distribution is crucial for the chemical industry's transition to net-zero emissions [155]. To further advance emission reduction efforts, enhanced data collection and coordination are needed, encouraging industry participation in tracking emissions and promoting data sharing among stakeholders.

Increase Research, Development, and Demonstration Investment: Developing financial mechanisms, such as green bonds and innovation funds, is essential to stimulate private investment in low-carbon technologies. Investments should focus on key areas like hydrogen production, CCS, and advanced recycling techniques to drive innovation and accelerate the transition to a sustainable future.

Create Markets for Low-Emission Products: Implement carbon contracts for difference to financially support the creation of near-zero- emission products, while encouraging public procurement to stimulate demand and accelerate the adoption of sustainable technologies [156].

Promote Energy Productivity: Establish mandatory energy audits and efficiency standards for industrial equipment, complemented by incentives for recycling and the use of efficient materials to promote sustainable practices [157].

Manage Transition of Existing Assets: Mandate upgrades of existing facilities to integrate advanced, low-emission technologies, reducing the risk

6.2. Outlook for the future of carbon neutrality in the chemical industry with the adoption of new and emerging technologies

The chemical sector, as a major carbon emitter, requires a comprehensive plan that includes legal, technological, and financial initiatives to reduce its carbon footprint. The chemical industry's decarbonization efforts are heavily influenced by national and regional regulatory frameworks, which vary in stringency, scope, and application. Governments implement carbon pricing, emissions trading, renewable energy incentives, and industrial efficiency mandates to drive emissions reductions, yet companies must navigate complex policy landscapes across different regions. The European Union, for example, has established the ETS and Carbon Border Adjustment Mechanism (CBAM) to manage energy-intensive industries like chemical production. The policies require companies to purchase carbon allowances, increasing carbon-intensive production costs and incentivizing low-carbon alternatives [145]. Similarly, the United States' Inflation Reduction Act (IRA) offers extensive funding for industrial decarbonization, including tax incentives for hydrogen production and carbon capture projects [146]. Such support makes technologies such as CCUS and green hydrogen more commercially viable for large-scale application. China's National Carbon Emission Trading Scheme (ETS), launched in 2021, is already the world's largest carbon trading system, covering thousands of factories, including chemical factories. The policy is set to reduce dependence on coal and enable cleaner production methods, accelerating the country's transition towards low-carbon manufacturing [147]. Japan has introduced its Green Growth Strategy and Carbon Neutral Industry Roadmap, providing funding for research and development in carbon-neutral materials, CCUS, and industrial electrification [148]. The strategies encourage innovation combined with financial support to lead industries towards cleaner practices. Given the diverse regulatory landscape, chemical companies must adopt strategic approaches to remain compliant and competitive. Implementing internal carbon pricing models can help firms anticipate future regulatory costs and guide investment decisions toward low-emission technologies [149]. Additionally, leveraging financial incentives, tax credits, and emissions trading revenues can offset the high capital expenditures required for decarbonization [150].

The following policy proposals from this study attempt to guide the industry toward sustainability: Comprehensive CO₂ Policies: Implementing economy-wide carbon pricing can drive emissions reductions while ensuring equity across industries, including chemicals [151]. Additionally, Carbon Border Adjustments, such as the EU's Carbon Border Adjustment Mechanism, are crucial to prevent carbon leakage and support global emission reduction efforts [152,153]. To further

of stranded assets and ensuring a smooth transition to sustainable practices [158].

The recommendation emphasizes that the chemical industry must adopt supply-side measures while adapting to reduced demand driven by the circular economy. It is crucial to accelerate technology development and scaling for raw material transition, finance, and supply chain data exchange on emissions and ingredients. Regulatory frameworks, such as carbon pricing and recycled content mandates, are essential. Collaboration across the supply chain must also be strengthened through downstream customer commitments to net-zero chemical products and the establishment of a global charter with transition principles [159].

6.3. Conclusion

This paper thoroughly investigates the many techniques that chemical firms are pursuing to attain carbon neutrality, emphasizing not only the difficulties but also the potential for considerable environmental and economic advantages. As the sector approaches a critical juncture, the shift to sustainable practices is not only desirable, but also required for its survival and competitiveness in a carbon-constrained future. The chemical industry has undergone a significant transformation, driven by advancements in technology, shifts to cleaner energy sources, and innovations in process management. These efforts not only contribute to combating climate change but also position companies to stay competitive in an evolving market. However, for these strategies to succeed, businesses must also tackle the challenges of scaling operations, cost efficiency, and seamless technology integration. A supportive regulatory and economic environment is essential to encourage innovation and accelerate the adoption of these critical technologies, ensuring long-term success and sustainability for the industry. Governments, investors, and industry leaders must work together to develop rules and incentives that stimulate significant investment in clean technologies. To keep current with changing technology and market trends, the industry must commit to continuous improvement and rigorous progress tracking, backed up by improved data gathering and analysis. Achieving the chemical industry's ambitious objective of carbon neutrality by 2050 will necessitate these ongoing efforts, establishing a strong example and paving the path for a sustainable and prosperous future. Our future work will focus on evaluating methods for assessing carbon reduction technologies, with an emphasis on modeling and optimization approaches. Additionally, this study acknowledges the need for deeper techno-economic and LCA of decarbonization technologies, ensuring a comprehensive evaluation of their environmental and economic impacts. These areas will be explored further in upcoming studies to enhance the effectiveness, feasibility, and scalability of industrial decarbonization strategies, supporting the long-term transition toward carbon neutrality in the chemical industry.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ladan Malehmirchegini reports financial support was provided by Mitsui Chemicals, INC.-Carbon Neutral Research Center (MCI-CNRC). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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